

HEPATO-BILIARY PANCREATIC SURGERY

Surgical Jaundice

INTRODUCTION / DEFINITIONS

- Jaundice (derived from French word 'jaune' for yellow or *icterus*, which is the Latin word for jaundice) is a yellowish staining of the skin, sclera and mucous membranes by deposition of bilirubin in these tissues.
- Jaundice indicates excessive levels of conjugated or unconjugated bilirubin in blood and is clinically apparent when bilirubin level exceeds 2 mg/dl.
- It is most apparent in natural sunlight.
- Causes of jaundice can be classified into pre-hepatic, hepatic or post-hepatic. Post-hepatic jaundice is also termed obstructive or surgical jaundice as this type is amenable to surgical treatment.

SURGICAL ANATOMY OF THE HEPATO-BILIARY SYSTEM

- The biliary system can be broadly divided into two components, the *intra-hepatic and the extra-hepatic tracts*.
- The secretory units of the liver (hepatocytes and biliary epithelial cells), the bile canaliculi, bile ductules and the intrahepatic bile ducts make up the *intra-hepatic* component of the biliary tree.
- The extra-hepatic bile ducts (right and left), the common hepatic duct (CHD), the cystic duct, the gallbladder (GB), and the common bile duct (CBD) constitute the *extra-hepatic* component of the biliary tree.
- The cystic and CHDs join to form the CBD which is approximately 8-0 cm in length and 0.4-0.8 cm in diameter. The CBD can be divided into three anatomical segments: *supra-duodenal, retro-duodenal, and intra-pancreatic* (Figure 1). It then enters the medial wall of the duodenum, courses tangentially through the submucosal layer for 1-2 cm, and terminates in the major papilla in the second portion of the duodenum (Figure 2). The distal portion of the duct is encircled by smooth muscle that forms the *sphincter of Oddi*. The CBD may enter the duodenum directly (25%) or join the pancreatic duct (75%) to form a common channel, termed the *ampulla of Vater*.

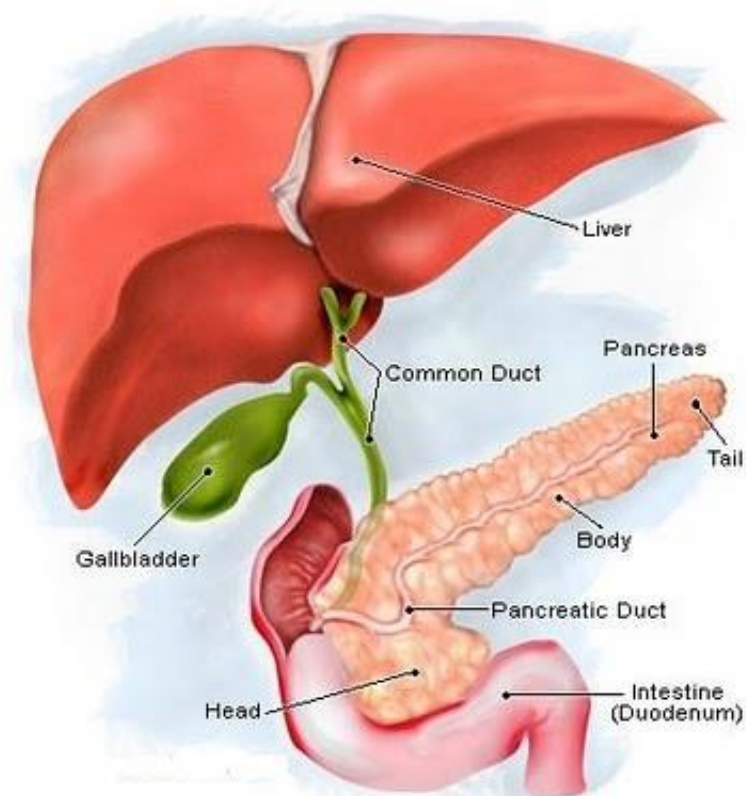


Figure 1. Anatomy of the common bile duct (CBD)

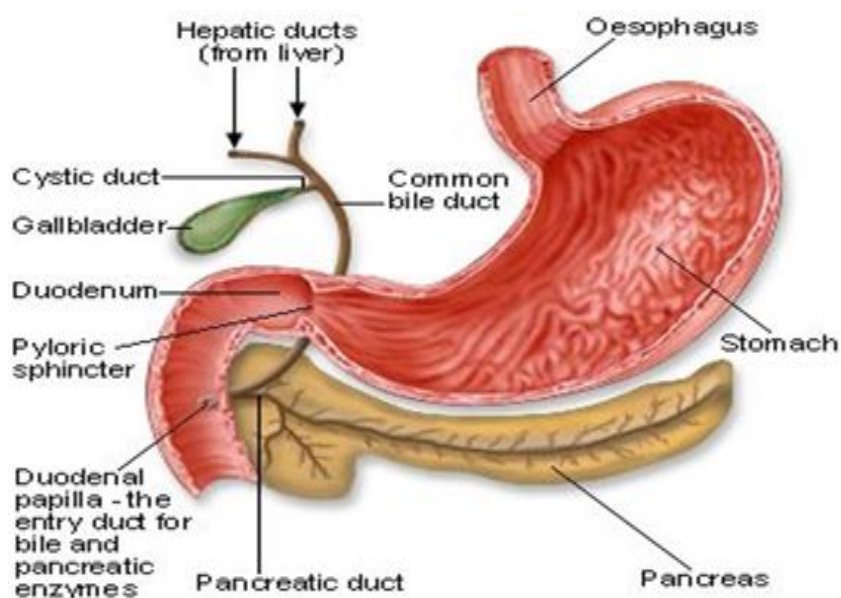


Figure 2. Anatomy of the biliary tree. The CBD enters the medial wall of the duodenum, courses tangentially through the submucosal layer for 1-2 cm, and terminates in the major papilla in the second portion of the duodenum

PHYSIOLOGY/BIOCHEMISTRY OF BILIRUBINE PRODUCTION AND TRANSPORT

- Bile is produced in the liver and contains bile salts, water, cholesterol, electrolytes, and bilirubin, which is a breakdown product of hemoglobin.
- The formation of bilirubin from heme is essential for mammalian life, because it provides the body with the main means of elimination of heme. Eighty percent of the circulating bilirubin is derived from heme of hemoglobin from senescent red blood cells (RBCs) destroyed in the reticuloendothelium of the bone marrow, spleen, and liver. Ten to twenty percent of the bilirubin comes from other sources such as myoglobin, cytochromes, and other heme-containing proteins processed in the liver.
- Initially, *heme* is oxidized to the green pigment *biliverdin*, which is then reduced to *bilirubin*, which is virtually insoluble in aqueous solutions. In blood it is reversibly, but tightly bound to plasma albumin.
- Hepatic uptake of bilirubin occurs with the dissociation of the albumin-bilirubin complex with subsequent translocation of bilirubin into the hepatocyte.
- In the hepatocytes, conjugation of bilirubin involves its esterification with glucuronic acid to form, first, a monoglucuronide, then a diglucuronide. The principal enzyme involved is uridine diphosphate (UDP)-glucuronyl transferase. Conjugation renders bilirubin water-soluble and is essential for its elimination from the body in bile and urine.
- *Conjugated* bilirubin is excreted in bile through the biliary and cystic ducts to enter the gallbladder, where it is stored; or it passes through the ampulla of Vater to enter the duodenum. Inside the intestines, some bilirubin is excreted in the stool as *stercobilin* (responsible for the dark color of stools), while the rest is metabolized by the gut flora into *urobilinogens* and then reabsorbed. The majority of the urobilinogens are filtered from the blood by the kidney and excreted in the urine. A small percentage of the urobilinogens are reabsorbed in the intestines and re-excreted into the bile through the entero-hepatic circulation

ETIOLOGY

Causes of surgical jaundice may be congenital or acquired as shown in Table 1.

Table 1: List of causes of surgical jaundice

CONGENITAL CAUSES	
1. Biliary atresia. 2. Cystic fibrosis Disease 3. Choledochal cysts	
ACQUIRED CAUSES	
1. Gallstone disease (Calcular obstructive jaundice):	a) Choledocholithiasis: - Primary: Formed inside the CBD - Secondary: Migrating from the GB b) Mirizzi Syndrome
2. Tumors (Malignant obstructive jaundice):	a) Primary tumors - The gallbladder - Bile ducts (cholangiocarcinoma) - Head of pancreas - Ampulla of Vater - Peri-ampullary carcinoma - Metastatic tumors b) Metastases deposited in the lymph nodes (LNs) at the porta-hepatis. Primaries of these metastatic deposits may be located in the lower esophagus, stomach, breast, or even the liver itself.
3. Trauma	a) Iatrogenic e.g. during cholecystectomy, liver resections, etc. b) Others e.g. stab wound, gunshot, etc
4. Strictures	a) Benign b) Malignant c) Primary sclerosing cholangitis

PATHOPHYSIOLOGY OF OBSTRUCTIVE JAUNDICE

- In obstructive jaundice, the pathophysiological effects reflect the absence of bile constituents in the intestines and their spillage into the systemic circulation.
- *Stools become pale* because less bilirubin reaches the intestine. Absence of bile salts can produce malabsorption, leading to steatorrhea and deficiencies of fat-soluble vitamins (particularly A, K, and D); vitamin K deficiency can reduce prothrombin levels and cause *coagulopathy*.

- Bilirubin retention produces mixed hyper-bilirubinemia (*jaundice*). Some conjugated bilirubin reaches the kidney and *darkens the urine*. High levels of circulating bile salts are associated with, but may not cause, *pruritus*.

CLINICAL FEATURES

- A good history, physical examination and diagnostic tests are the requisites for the evaluation of the jaundiced patient.
- Jaundice, dark urine (liquorice colored), pale stools and generalized pruritus are the hallmark of obstructive jaundice.
- The main aim of history-taking and physical examination is the differentiation between calcular (benign) and malignant obstructive jaundice (Table 2).
- History of fever, biliary colic and intermittent jaundice may be suggestive of calcular obstructive jaundice.
- Weight loss, abdominal mass, pain radiating to the back and progressively deepening jaundice may be suggestive of malignant obstructive jaundice.
- Deep jaundice (with a greenish hue) that appears to fluctuate *in intensity* may be due to a peri-ampullary cancer.

Table 2: Differences between calcular and malignant obstructive jaundice

Features	Calcular	Malignant
1. History	Long (months or years)	Short (days or weeks)
2. Symptoms - Dyspepsia - Biliary colic - Anorexia - Weight loss	Fatty food intolerance Present Absent Absent	Usually absent Absent Present Present
3. Onset of jaundice	After a heavy meal and an attack of typical biliary colic (<i>Painful Jaundice</i>)	Accidental discovery i.e. jaundice usually told by others (<i>Painless Jaundice</i>)
4. Course of jaundice	Intermittent	Progressive
5. Cholangitis	Present	Absent
6. Previous attacks	±	Never

Cholangitis

- *Definition:* It means infection of the biliary system.
- *Clinical presentation:* It can present as either:
 - a) *Charcot's triad:* right upper quadrant pain, jaundice, and fever with rigors.
 - b) *Reynolds pentad:* right upper quadrant pain, jaundice, and fever with rigors, hypotension and an altered mental status.

Courvoisier's Law

Courvoisier's law

In a jaundiced patient, if gallbladder is palpable, it is unlikely to be due to gallstones

Explanation of the law

Malignant obstruction is usually a complete obstruction, which leads to chronically elevated intra-ductal pressures. This contrasts with obstructions caused by gallstones; stones usually cause only partial obstructions (related to a “ball-valve” action of the stone) leading to less consistent intra-ductal pressure elevations and less gallbladder dilatation.

Exceptions of the law

1. A malignant obstruction above the cystic duct will *not* cause GB dilatation. Therefore, absence of a palpable GB in a jaundiced patient does exclude a malignant cause
2. Rarely, a stone may get impacted at the cystic duct while another obstructs the CBD leading to a palpable mucocele of the GB and obstructive jaundice. Nevertheless, this case should never underscore the fact that for patients presenting with jaundice and an enlarged palpable GB the possibility of a neoplastic process involving the head of pancreas or the ampulla of Vater should be strongly considered.

INVESTIGATIONS

Biochemistry / Hematology

- Elevated serum bilirubin level with a preponderance of the conjugated fraction is the rule.
- The serum gamma glutamyl trans-peptidase (GGT) level is also raised in cholestasis.
- The alkaline phosphatase may be elevated up to ten times normal.
- Elevated white cell count (WBC) may be present in cholangitis.
- Tumor markers like CA 19-9, CEA and CA-125 are usually elevated in pancreatic cancers, cholangiocarcinoma and peri-ampullary cancers, but they are non-specific and may be elevated in other benign diseases of the hepatobiliary tree.

Imaging

Goals of Imaging

1. To confirm the presence of an extra-hepatic obstruction (i.e. to verify that the jaundice is indeed post-hepatic rather than hepatic),
2. To determine the level of the obstruction,
3. To identify the specific cause of the obstruction, and
4. To provide complementary information relating to the underlying diagnosis (e.g. staging information in cases of malignancy)

Ultrasonography (US)

It is the most important initial tool of investigation as it can provide sure evidence of biliary tree obstruction; namely *intra-hepatic bile duct dilatation*. In addition it can provide data regarding:

- The size of the extra-hepatic biliary tree
- The level of the obstruction,
- The cause and additional information related to the disease (e.g. GB stones, hepatic metastases, hepatic parenchymal change).

Computed Tomography (CT) Scan of the Abdomen

It is very useful for staging of pancreatic and biliary malignancies. It provides excellent visualization of the tumor and its relation to surrounding structures, as well as visualization of the liver and lymphadenopathies.

Endoscopic Ultrasound (EUS)

- With regard to the biliary system, EUS is useful for the detection and staging of ampullary tumors, evaluation of benign and malignant bile-duct strictures and finally in evaluating relationships to vascular structures.
- EUS enables the aspiration of cysts and biopsy of solid lesions.

Direct Visualization of the Biliary Tree

1. Endoscopic retrograde cholangio-pancreatography (ERCP)
2. Percutaneous trans-hepatic cholangiography (PTC):
PTC provides direct visualization of the level of obstruction. Both ERCP and PTC are invasive and may be associated with complications such as cholangitis, biliary leakage, pancreatitis, and bleeding.
3. Magnetic resonance cholangio-pancreatography (MRCP)
It is a non-invasive technique for visualization of the biliary and pancreatic ductal system.

TREATMENT OF OBSTRUCTIVE JAUNDICE

Treatment of obstructive jaundice is treatment of the cause

Surgically Correctable Jaundice

The term surgically correctable jaundice refers to a broader spectrum of diseases where surgery can relieve jaundice.

1. *Pre-hepatic (hemolytic) jaundice:*

Hereditary Spherocytosis:

- Hereditary spherocytosis is a disorder of the RBC membrane that leads to sequestration and destruction of the spherocytic RBC in the spleen resulting in hemolytic anemia.
- Splenectomy is curative and serves as the sole mode of therapy.

2. *Hepatic jaundice*

- End-stage liver disease requiring *liver transplantation*

3. *Post-hepatic (obstructive) jaundice*

Surgical Diseases of the Gall Bladder

CHOLELITHIASIS

The spectrum of manifestation of gallstones is very wide. Figure 1 illustrates the most common of these manifestations.

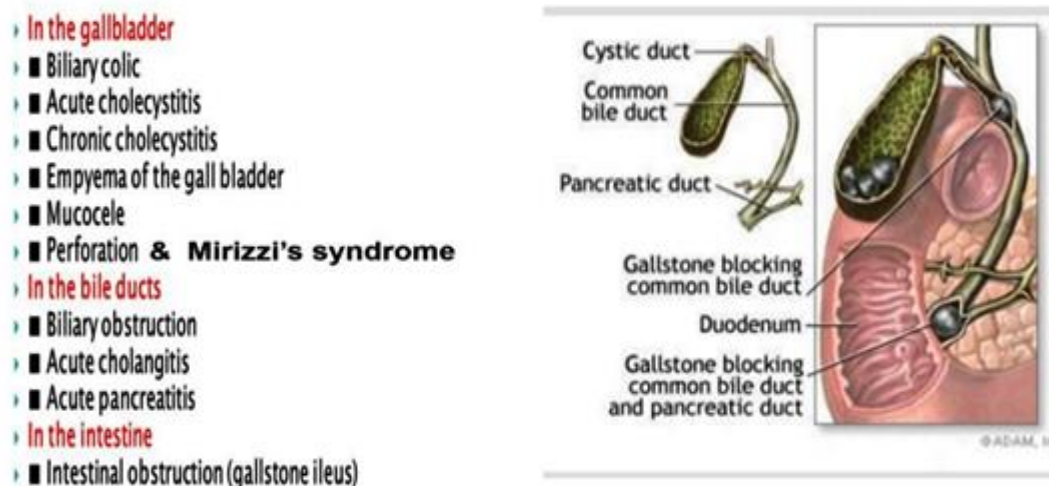


Figure 1: Spectrum of gallstone manifestations and complications

Asymptomatic Gallstones

Clinical Presentation

Asymptomatic gallstones are usually discovered on routine imaging studies or incidentally at laparotomy for unrelated problems.

Management

- There is no role for prophylactic cholecystectomy in most patients with asymptomatic gallstones.
- Prophylactic cholecystectomy may be warranted in patients with asymptomatic gallstones who have other risk factors for gallbladder cancer as outlined later.

- Children with gallstones have a relative indication for cholecystectomy due to the general difficulty of declaring and interpreting symptoms in this population.
- Management of gallstones discovered at laparotomy remains controversial because the literature is conflicting with regard to the incidence of biliary symptoms after surgery in patients in whom the gallbladder is not removed.

Symptomatic Gallstones

Clinical Presentation

- *Biliary colic* is the main symptom and is initiated by impaction of a gallstone in the outlet of the gall bladder (GB). An attack has characteristic periodicity, location, and timing. The pain comes in waves lasting 30 minutes to several hours and is typically situated in the epigastrium or right upper quadrant, occasionally with concomitant back or left upper quadrant symptoms. The pain is commonly severe after a meal with such intensity as to wake-up the patient from sleep.
- *Other symptoms* include nausea and vomiting.

Diagnosis

- Physical signs include mild right upper quadrant tenderness, although there may be few abdominal findings during an attack. If jaundice is present, another cause should be sought.
- Ultrasound (US) diagnosis is based on the presence of echogenic structures having posterior acoustic shadows (stones). There is usually little or no associated GB wall thickening or other evidence of cholecystitis. The bile ducts must be assessed for evidence of dilatation or choledocholithiasis (bile duct stones).

Treatment

- Laparoscopic cholecystectomy (LC) is the appropriate treatment of patients with symptomatic gallstones, as described below.

Acute Calculous Cholecystitis

It is initiated by obstruction of the cystic duct by an impacted gallstone. Persistence of stone impaction leads to inflammation of the gallbladder.

Diagnosis

- Diagnosis can be made by a combination of local and systemic signs of inflammation, correlated with imaging findings.
- Local inflammatory signs include right upper quadrant pain and tenderness as well as *Murphy sign*, which is inspiratory arrest during deep palpation of the right upper quadrant.
- Systemic signs include fever, leukocytosis, and an elevated C-reactive protein (CRP) level. Mild jaundice may be present, but severe jaundice is rare and suggests the presence of common bile duct (CBD) stones, cholangitis, or obstruction of the CBD caused by external compression from Mirizzi syndrome.

- Complications, such as gangrene, perforation, or cholangitis, are suggested by moderate leukocytosis ($>20,000$ cells/ μL).
- Liver function tests (LFTs), including serum bilirubin, alkaline phosphatase, alanine transaminase (ALT), aspartate transaminase (AST), and serum amylase, may also be abnormal.
- Ultrasonography (US) may reveal GB wall thickening, peri-cholecystic fluid, and a sonographic Murphy sign (tenderness over the GB when compressed by the US probe).
- Computed tomographic (CT) scanning is now frequently performed to evaluate the patient with acute abdominal pain. It can demonstrate gallstones, although it is less sensitive for these than US. Other signs of acute cholecystitis on CT scan include GB wall thickening, peri-cholecystic fluid, edema, and emphysematous changes.
- Radionuclide cholescintigraphy can be useful as an adjunct in the diagnosis of acute cholecystitis. Scintigraphic scanning with (HIDA) enables visualization of the biliary system. The radionuclide is concentrated and secreted by the liver, allowing visualization of the bile ducts and the GB normally within 30 minutes. Since the test depends on hepatic excretion of bile, it may *not* be useful in jaundiced patients. Non-filling of the GB after 4 hours is diagnostic of acute cholecystitis. Although its sensitivity and specificity are higher than US, its expense and total study duration limit it from being the first imaging choice.

Management

- Initial steps for patients with acute cholecystitis include hospitalization, intravenous (IV) fluid resuscitation, and parenteral antibiotics.
- In Mild/ responsive cases for medical treatment
 1. Early approach: within first 72 hours of the attack onset
 2. Delayed approach: after 6 weeks of treatment
- In Complicated cases/ or failed medical treatment:
 1. Emergency cholecystectomy
 2. Percutaneous Cholecystostomy

Several prospective, randomized trials have compared early *versus* delayed (6 weeks) LC for acute cholecystitis. Recent meta-analyses of the existing literature showed no significant differences in early *versus* delayed procedures with regard to mortality, conversion rate, bile duct injury, and peri-operative complications.

Choledocholithiasis

Choledocholithiasis is generally due to gallstones that originate in the GB and pass through the cystic duct into the CBD. Stones rarely originate in the hepatic or common ducts in Western countries, but intra-hepatic stones are more common in Asia.

Diagnosis

- The most common manifestation of uncomplicated choledocholithiasis is *jaundice*, with bilirubin levels typically between 3-10 mg/dL.
- *Cholangitis* is often caused by choledocholithiasis.

- Biliary colic is common.
- Ultrasonography usually demonstrates GB stones and bile duct dilatation. Because of obscuring gas in the duodenum, ductal stones are visible in only about 50% of cases.
- Diagnosis may be confirmed by magnetic resonance cholangio-pancreatography (MRCP), endoscopic retrograde cholangio-pancreatography (ERCP) or percutaneous trans-hepatic cholangiography (PTC), which can opacify the biliary tree and demonstrate the intra-ductal stones.
- Occasionally, diagnosis of choledocholithiasis is confirmed by intra-operative cholangiography (IOC) at the time of cholecystectomy.

Management

Management depend on the clinical situation as follows:

- Patients with low risk of choledocholithiasis have no evidence of LFT derangements, jaundice, cholangitis, pancreatitis, or imaging evidence of CBD stones. In these patients, standard management consists of LC.
- Patients with high risk of choledocholithiasis have US evidence of CBD stone, bilirubin levels >4 mg/dL, clinical ascending cholangitis, or dilated CBD on US with a bilirubin >1.8 mg/dL. In these patients, ERCP is recommended prior to interval LC. ERCP with sphincterotomy carries a <1% risk of mortality and a 5-10% risk of morbidity, principally acute pancreatitis.
- In cases with ectatic CBD (>1.5cm in diameter); biliary enteric bypass is needed.

Biliary Pancreatitis

- Biliary Pancreatitis is caused by blockage of pancreatic secretions by passage of a gallstone into the common biliary pancreatic channel. The greatest risk is carried by small (<2 mm) stones.
- Once the acute episode of pancreatitis has resolved, the GB should be removed during the same admission to avoid recurrent pancreatitis.
- A longer delay may be justified in patients who have had severe pancreatitis and in whom local inflammation or systemic illness contraindicates surgery.

Gallstone Ileus

- Gallstone ileus is a small bowel obstruction caused by a gallstone, an uncommon complication resulting from a gallstone eroding through the GB into the adjacent bowel (usually duodenum). The stone migrates until it lodges in the narrowest portion of the small bowel, just proximal to the ileo-cecal valve.
- Patients present with symptoms of bowel obstruction and ascending cholangitis from the cholecysto-enteric fistula.
- Treatment is exploratory laparotomy and removal of the obstructing gallstone by milking it back to an enterotomy made in healthy intestine. The entire bowel should be searched diligently for other stones, and cholecystectomy with closure of the fistula should be performed if the patient is stable and the inflammation is *not* too severe.

Laparoscopic Cholecystectomy (LC)

Indications

- Laparoscopic cholecystectomy is the treatment of choice for all cases of *symptomatic* gallstone disease.
- In case of *asymptomatic* gallstones, LC is indicated in the following conditions:
 1. Calcified GB
 2. Gallstone > 3cm in diameter
 3. Sickle cell disease
 4. Diabetic patients (relative indication)
 5. Young patients (relative indication)
- The complication rate of LC is low and is available for 95% of patients, with excellent recovery and return-to-work times.

Contraindications

- Concomitant diseases that prevent use of a general anesthetic.
- Patient's refusal of open cholecystectomy should urgent conversion be required.

Technique

- Because mis-identification of the cystic duct is the commonest cause of biliary injury, the surgeon must use a technique to provide conclusive identification of the cystic duct and artery in *Calot's triangle* and *hepato-cystic triangle* is displayed in Figure 2.

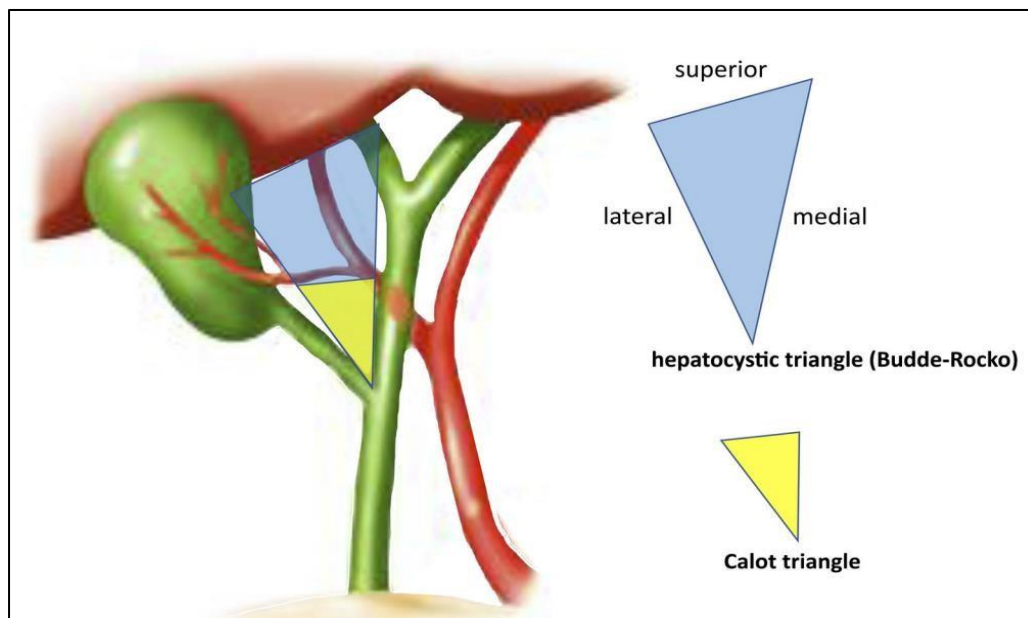


Figure 2: Hepatocystic triangle and triangle of Calot. Hepatocystic triangle (blue): Upper boundary of hepatocystic triangle is the inferior border of liver. Laterally, the cystic duct and neck of the gallbladder, medially, the common hepatic duct. Triangle of Calot (yellow): Upper boundary is the cystic artery. Lateral the cystic duct. Medial the CBD

- In the critical view of safety technique, the triangle of Calot is dissected free of fat, fibrous, and areolar tissue. Importantly, the lower end of the GB must be dissected off the liver bed. A complete dissection demonstrates only two structures (i.e. the cystic duct and cystic artery) entering the GB, constituting the critical view of safety. To assist with anatomical definition, IOC may be used, especially whenever the critical view is not achieved.

Complications

- LC appears to be associated with a higher incidence (2.5/1,000) of major bile duct injury than open cholecystectomy. In addition, there are also risks of injury of other structures, including the hepatic artery and the bowel.
- Spilled and retained gallstones can be the source of infrequent, but serious long-term complications such as abscess and fistula formation.
- Factors associated with an increased rate of conversion to an open procedure include emergent cholecystectomy, male sex, age >60 years, obesity, severe GB inflammation, choledocholithiasis, and prior upper abdominal surgery.

Open Cholecystectomy

- Open cholecystectomy is performed in patients (1) who have contraindications to LC, (2) who require conversion from LC because of inability to complete the laparoscopic procedure, or (3) who are undergoing laparotomy for another operation (e.g. pancreaticoduodenectomy).
- *Conversion* to an open operation in the face of a difficult laparoscopic procedure should never be viewed as a surgical failure or complication, but rather as a way to avoid potential injury to the patient.
- A *partial cholecystectomy* is advocated when the ductal and vascular structures in the triangle of Calot cannot be safely identified in the setting of severe acute inflammation. In this situation, the fundus is opened and stones are extracted. The anterior wall of the GB is excised, leaving the posterior wall with the liver bed. The mucosa is then fulgurated, and a drain is left near the infundibulum.

Summary

Diagnosis of the different manifestations of gallstones is summarized in Table 1.

Table 1: Diagnosis of the different manifestations of gallstones

Complication	History	Examination	Blood tests
Biliary Colic	- Intermittent RUQ/epigastric pain (minutes/hours) into back or right shoulder - N&V	- Tender RUQ - No peritonism - Murphy's - - Apyrexial, HR and BP (N)	- WCC (N) CRP (N) - LFT (N)
Acute Cholecystitis	- Constant RUQ pain into back or right shoulder - N&V - Feverish	- Tender RUQ - Peritonism RUQ (guarding/rebound) - Murphy's + - Pyrexia, HR (!)	- WCC and CRP (!) - LFT (N or mildly (!))
Empyema	- Constant RUQ pain into back or right shoulder - N&V - Feverish	- Tender RUQ - Peritonism RUQ - Murphy's + - Pyrexia, HR (!), BP (→ or !) - More septic than acute cholecystitis	- WCC and CRP (!) - LFT (N or mildly (!))
Obstructive Jaundice	- Yellow discolouration - Pale stool, dark urine - painless or associated with mild RUQ pain	- Jaundiced - Non-tender or minimally tender RUQ - No peritonism - Murphy's - - Apyrexial, HR and BP (N)	- WCC and CRP (N) - LFT: obstructive pattern bili (!), ALP (!), GGT (!), ALT/AST (→) - INR (→ or !)
Ascending Cholangitis	Becks triad - RUQ pain (constant) - Jaundice - Rigors	- Jaundiced - Tender RUQ - Peritonism RUQ - Spiking high pyrexia (38-39) - HR (!), BP (→ or !) - Can develop septic shock	- WCC and CRP (!) - LFT : obstructive pattern bili (!), ALP (!), GGT (!), ALT/AST (→) - INR (→ or !)
Acute Pancreatitis	- Severe upper abdominal pain (constant) into back - Profuse vomiting	- Tender upper abdomen - Upper abdominal or generalised peritonism - Usually apyrexial, HR (!), BP (→ or !)	- WCC and CRP (!) - LFT: (N) if passed stone or obstructive pattern if stone still in CBD - Amylase (!) - INR/APTT (N) or (!) if DIC
Gallstone Ileus	- 4 cardinal features of SBO	- distended tympanic abdomen - hyperactive/tinkling bowel sounds	

BENIGN STRICTURES AND BILE DUCT INJURIES

Etiology

- Benign strictures occur in association with a number of conditions, including (1) chronic pancreatitis, (2) choledocholithiasis, (3) primary sclerosing cholangitis (PSC), (4) prior

hepatic transplantation, (5) trauma, or (4) iatrogenic injury after instrumentation or surgery. LC is the leading cause of *iatrogenic* bile duct injuries and subsequent benign strictures.

- Biliary malignancy may masquerade as a benign stricture. Attempts to differentiate between the two should be made prior to surgery because the patient may not be a candidate for a curative resection if advanced cancer is found.

Bismuth Classification

Bismuth classification for biliary strictures is based on transections or occlusions at various levels of the CBD as shown in Figure 3.

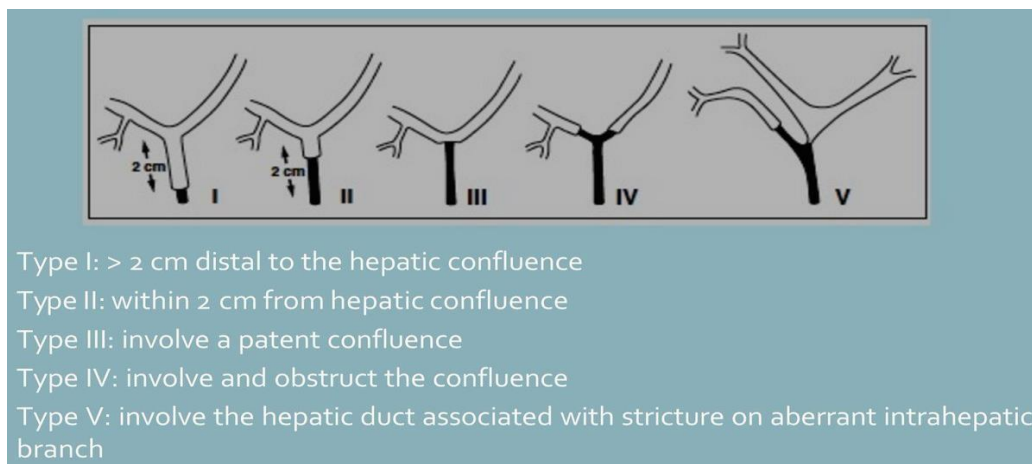


Figure 3: Bismuth classification of biliary strictures (Types I-V)

Clinical Presentation

- Clinical Presentation depends on the type of injury. Approximately 25% of major bile duct injuries are recognized at the time of the initial procedure. Intra-operative signs of a major ductal injury include unexpected bile leakage, abnormal IOC, and delayed recognition of the anatomy after transection of important structures.
- If an injury is not recognized intra-operatively, the patient usually presents with symptoms within 1 week and within 3-4 weeks after the initial procedure.
- Patients with a bile leak often present with right upper quadrant pain, fever, and sepsis secondary to biloma and may have bile drainage from a surgical incision.
- Patients with occlusion of the CBD without a bile leak present with jaundice. Occasionally, a delayed presentation of months or years is seen, especially in case of ischemic biliary stricture.

Diagnosis

- Ultrasound (US) abdomen is usually the initial imaging, which can detect collections and /or biliary dilatation.
- Axial imaging with CT scan or magnetic resonance imaging (MRI) is useful for detecting abdominal bile collections that require percutaneous drainage.

- MRCP with angiography is now often the initial imaging test of choice because of its ability to define the biliary anatomy and any associated vascular injuries.
- On-going bile leaks can also be diagnosed by HIDA scan.
- ERCP is for diagnostic and therapeutic purposes, such as *biliary stent* deployment.
- In the case of occlusion of the CBD, PTC can demonstrate the biliary anatomy distal to the occlusion and be used for decompression of the biliary tree.

Management

Management depends on the clinical presentation.

- If the injury is identified at the time of the initial procedure, the surgeon should proceed directly to open exploration and repair, only if trained with complex techniques in hepato-biliary surgery, or to control life-threatening hemorrhage. Otherwise, the patient should be resuscitated, a drain should be placed in the right upper quadrant, and the patient immediately referred to a hepato-biliary specialist.
- Some simpler injuries can be successfully managed with ERCP and sphincterotomy and stenting. Occlusive lesions, usually with clip occlusion, require decompression of the proximal system via PTC.
- Ideally, in delayed diagnosis, temporization for at least 8 weeks can allow the acute inflammation to resolve. Control of sepsis, percutaneous drainage, and adequate nutrition should be optimized before definitive repair. In addition, if there is a concern about a concomitant vascular injury, definitive repair should be delayed to identify areas of ductal ischemia more easily, which should not be incorporated in the repair.
- *Operative repair* is best achieved by means of a Roux-en-Y hepatico-jejunostomy after debridement of the bile duct to viable tissue. All bile ducts must be accounted for, and an adequate blood supply must be apparent for each. A tension-free mucosa-to-mucosa anastomosis constructed with fine absorbable suture is desired. Excellent long-term outcomes have been described with anastomotic stricture is the most common, yet infrequent, complication.

BENIGN STRICTURES AND BILE DUCT INJURIES

Clinical Presentation

- Acute cholangitis is a potentially life-threatening bacterial infection of the biliary tree typically associated with obstruction of the ductal system.
- Although acute cholangitis is often associated with choledocholithiasis, other causes include benign and malignant strictures of the bile ducts or at biliary-enteric anastomoses, parasites, and indwelling tubes or stents, but almost always this occurs with incomplete biliary decompression (e.g. stent occlusion).
- ERCP without concomitant stenting in the presence of a stricture may lead to cholangitis above the stricture. Therefore, patients should routinely be pre-treated with antibiotics in case a stent cannot be placed.

Diagnosis

- Patients present with a spectrum of disease severity, ranging from subclinical illness to acute toxic cholangitis.
- Fever is present in >90% of patients. Charcot triad (fever, jaundice, and right upper quadrant pain) is present in only 50-70% of patients, and Reynold's pentad (Charcot triad with hemodynamic instability and mental status changes) in <10% of patients, mostly in the elderly and those with a septic course.
- Laboratory data may demonstrate leukocytosis and abnormalities in LFTs.
- Ultrasonography or CT scan can reveal gallstones and biliary dilatation, but definitive diagnosis is made by ERCP or PTC. These studies are both diagnostic and therapeutic because they demonstrate the level of obstruction and allow culture of bile, removal of stones or indwelling foreign bodies, and placement of drainage catheters if necessary.

Treatment

- Initial management of cholangitis includes IV antibiotics appropriate for the coverage of the most common Gram-negative aerobic and anaerobic organisms.
- In patients with acute toxic cholangitis or in patients who fail to respond to antibiotic therapy, emergent decompression of the biliary tree via ERCP or PTC is required.
- Cholangitis in patients with indwelling tubes or stents generally requires stent removal and replacement.
- Definitive operative therapy for benign or malignant biliary tract strictures should be deferred until a later date.

GALL BLADDER CANCER

Pathology

- Histologically, nearly all gallbladder cancers are adenocarcinomas, and concomitant cholecystitis is frequently present.
- Tumors spread primarily by direct extension into liver segments IV and V but also via lymphatics along the cystic duct to the CBD.

Presentation

- Gall bladder cancer is the most common malignancy of the biliary tract. It is more aggressive than cholangio-carcinoma and has a poor prognosis with median survival of 5-8 months.
- Because of its generally advanced stage at presentation, only a small percentage of patients with a pre-operative diagnosis of GB cancer are resectable for potential cure.
- Polyps 1.5 cm or greater in diameter have a 46-70% prevalence of cancer, whereas in those smaller than 1 cm, the risk of malignancy is <5%.
- Malignant polyps also tend to be sessile in nature and echogenic on ultrasound.

- Prophylactic cholecystectomy should be considered for polyps >1 cm in size or meeting morphologic criteria.
- Other risk factors include porcelain gallbladder, PSC, and anomalous junction of the pancreato-biliary duct.

Diagnosis

- Approximately one-third of these tumors are diagnosed incidentally during cholecystectomy, found in 0.3% to 1% of all cholecystectomy specimens.
- Symptoms of stage I and II GB cancer are often directly caused by gallstones rather than the cancer, whereas stage III and IV cancers present with weight loss and symptoms typical of CBD obstruction.
- Suggestive ultrasound findings include thickening or irregularity of the GB, a polypoid mass, or diffuse wall calcification indicative of porcelain GB.

Treatment

- Mucosal disease confined to the GB wall (Tis and T1a tumors) is often identified after routine LC. Because the overall 5-year survival rate is as high as 80%, cholecystectomy alone with negative resection margins, including the cystic duct margin, is adequate therapy.
- Patients with a pre-operative suspicion of GB cancer should undergo open cholecystectomy because port site recurrences and late peritoneal metastases (associated with bile spillage) have been reported even with in situ disease.
- T2 tumors have invaded the muscularis and may be treated by radical cholecystectomy that includes the GB, the GB bed of the liver, and the hepato-duodenal ligament, para-duodenal, peri-pancreatic, hepatic artery, and celiac lymph nodes (LNs). Presence of LN metastases or extension of disease beyond the GB wall into local organs (T3–T4 disease) requires more radical resection. Depending on the extent of local invasion, extirpation may range from wedge resection of the liver adjacent to the GB bed to resection of 75% of the liver.
- Improvement in survival has been demonstrated after radical resection.
- Because of the aggressive nature of this malignancy, adjuvant chemo-radiation is often recommended, but little proof of efficacy is available. Most GB cancers have invaded adjacent organs, extended into the porta hepatis, or distantly metastasized before clinical diagnosis. Extensive liver involvement or dis-contiguous metastases preclude surgical resection. Jaundice may be palliated by percutaneous or endoscopically placed biliary stents. Duodenal obstruction can be surgically bypassed if present.

Acute Pancreatitis

DEFINITION AND EPIDEMIOLOGY

- Acute pancreatitis describes an acute inflammatory process of the pancreas that can range from mild interstitial pancreatitis to severe pancreatitis with pancreatic necrosis and concomitant multi-organ failure. Acute pancreatitis is typically rapid in onset and most commonly encountered in its mild form.
- The increased frequency of acute pancreatitis may be due to rising incidence of obesity, a risk factor for the development of gallstones and, by extension, gallstone pancreatitis.
- Acute pancreatitis confers a heavy financial burden on the health care system and significant physiological stress on the patient. The average length of hospital stay for a patient with acute pancreatitis is approximately 5-6 days. In addition, acute pancreatitis may be accompanied by life-threatening complications as well as significant morbidity and mortality.

ETIOLOGY

- The two *most common causes* of acute pancreatitis are:
 1. Cholelithiasis / choledocholithiasis
 2. Alcohol consumption (definitions vary as does duration with consumption between 50-80 g or 4–7 drinks/day) with frequency estimates of 40% and 30%, respectively.
- *Other etiologies* are hyper-triglyceridemia (>1000 mg/dL), medications, trauma, infections, iatrogenic (surgical or post-ERCP), genes, anatomy (pancreatic divisum, and sphincter of Oddi dysfunction, which remain controversial), as well as, autoimmunity.

Gallstones and Biliary Pancreatitis

- Biliary pancreatitis, synonymous with gallstone pancreatitis, is a form of acute pancreatitis caused by the passage of gallstones through the cystic duct and into the distal common bile duct (CBD) where they can obstruct the biliary and pancreatic ducts. Pancreatic ductal obstruction is felt to be the inciting event in gallstone pancreatitis.
- Incidence is highest in patients with small gallstones, or microlithiasis, as these stones are more likely to escape the gallbladder (GB) and transit the cystic duct to reach the CBD. Large stones are more likely to be retained in the GB. Gallstone pancreatitis typically presents and tended to have a more benign clinical course with infrequent ICU admissions.

- Obesity is a risk factor for the development of gallstone pancreatitis and may increase the risk of development of severe pancreatitis, including pancreatic necrosis. Obesity also increases the risk of developing local complications such as pancreatic fluid collections. However, obesity is *not* associated with increased mortality in acute pancreatitis.
- Other risk factors associated with the development of gallstone acute pancreatitis include pregnancy, elevated alanine aminotransferase (ALT), advancing age, weight gain, female gender, and rapid weight loss. Unsaturated fats, coffee, and moderate alcohol consumption appear to reduce the risk of developing gallstones and thus may reduce the risk for gallstone acute pancreatitis.

Alcoholic Pancreatitis

- Alcohol is a well-known precipitant of acute pancreatitis, although the incidence of acute pancreatitis in heavy alcohol consumers is not more than 2-3% per year, suggesting that there are as yet undetermined environmental or genetic factors that influence the development of acute pancreatitis in this population.
- Currently, 17.6 million Americans have an alcohol use disorder, and some data suggest that the incidence of alcoholic acute pancreatitis is on the rise. Furthermore, alcoholic acute pancreatitis has the highest associated risk of overall mortality, with the odds of death increased 90% as compared with biliary pancreatitis, possibly due to poor baseline nutrition.

Iatrogenic Pancreatitis

- Iatrogenic pancreatitis most commonly occurs following endoscopic retrograde cholangio-pancreatography (ERCP). Post-ERCP pancreatitis has an incidence of approximately 3.5%.
- Young age, biliary sphincter balloon dilatation in intact papilla, pancreatic duct contrast injection, normal bilirubin, precut sphincterotomy or pancreatic sphincterotomy, and suspected sphincter of Oddi dysfunction are risk factors for post-ERCP pancreatitis.
- Pancreatitis may also occur following abdominal surgery, cardiac surgery, liver biopsy, and abdominal procedures performed by interventional radiologists. It can also be caused by retained intra-abdominal foreign bodies as well as iatrogenic hypercalcemia due to total parental nutrition, among other causes.

Drug-induced Pancreatitis

- Medications thought to induce acute pancreatitis have been classified on the level of evidence to support the association.
- *Class I medications* are defined as those where recurrence of acute pancreatitis was confirmed upon re-challenging. Class I is further subdivided into 1a (other causes of pancreatitis ruled out) and 1b (alternative etiologies not ruled out).
- *Class II medications* do not meet strict criteria for class 1 but exhibit a consistent latency period in a preponderance of reported cases.

- *Class III and IV medications* refer to those in which two or one published case report of medication- induced pancreatitis has been reported, respectively.
- Medications implicated in the development of drug-induced pancreatitis are summarized in Table 1.

Table 1: Drugs (medications) implicated in development of pancreatitis

Class	Drug(s)
- Antimicrobials	- Tetracycline, sulfonamide, pentamidine, didanosine, metronidazole
- Anti-convulsants	- Valproic acid
- Diuretics	- Furosemide, thiazides
- Immunosuppressants	- Azathioprine, 6-mercaptopurine
- Non-steroidal anti-inflammatory drugs	- Sulindac, salicylates
- Anti-proliferative drugs	- Tamoxifen, L-asparaginase
- Others	- Estrogen

Hyper-triglyceridemia and Hyper-calcemia

- Elevated triglyceride serum levels, typically exceeding 500 mg/dL, can be seen in various conditions, including poorly controlled type-2 diabetes mellitus (DM), obesity, alcoholism, third trimester pregnancy, renal disease, hypothyroidism, and familial hypertriglyceridemia.
- Patients should have fasting triglyceride levels checked after their pancreatitis has resolved before diagnosing hyper-triglyceridemia, as serum triglycerides can be artificially elevated during an episode of acute pancreatitis.
- Hypercalcemia is also a recognized etiology of acute pancreatitis. Hypercalcemia can be associated with a malignancy (often in the setting of bony metastases or multiple myeloma), total parenteral nutrition, sarcoidosis, vitamin D toxicity, and infusions of peri-operative high-dose calcium during cardio-pulmonary bypass. If muscular/myopathic, urological, or nervous system symptoms co-exist with acute pancreatitis, patients should be evaluated for hyperparathyroidism (HPT).

Autoimmune Pancreatitis

- It is a predominantly lymphocytic inflammatory process that results in eventual organ fibrosis and dysfunction.
- While many diagnostic criteria have been proposed, the modified Japan Pancreas Society Criteria require a combination of typical imaging (CT, MRCP, or ERCP) and either serology (IgG4 or IgG totals, etc.) or pancreatobiliary / extra-intestinal findings (sialadenitis, nephritis, or IgG4 pneumonitis) for diagnosis.

Infectious Causes

- Infectious causes of acute pancreatitis are rare and have mostly been described in case reports.
- *Parasitic*: the most common are Toxoplasma, Cryptosporidium, and Ascaris.
- *Viral*: examples include mumps, Coxsackie virus, hepatitis B, cytomegalovirus (CMV), and varicella-zoster virus.
- *Bacterial*: examples include Mycoplasma, Legionella, Leptospira, and Salmonella.
- *Fungal*: Aspergillus is the only fungus that has been shown to cause acute pancreatitis.

Inherited Forms

- Mutations, up-regulation, and genetic variants in several genes have been implicated in acute pancreatitis, namely, the trypsinogen gene (PRSS1) and trypsin inhibitor (SPINK1), cystic fibrosis trans-membrane regulator (CFTR) variants, and endothelial ion/water channel CLDN2 risk allele.

Other Causes

- *Neoplasms*: Rarely, tumors such as ampullary cancer or intra-ductal papillary mucinous neoplasm of the pancreas can cause acute pancreatitis.
- *Congenital malformations*, including pancreas divisum, annular pancreas, and anomalous pancreatico-biliary union, among others, have also been implicated.

Idiopathic Acute Pancreatitis

- Between 10% and 30% of cases of acute pancreatitis may be idiopathic in nature.
- It may be explained by undetected micro-lithiasis, unrecognized drug-induced pancreatitis, or sphincter of Oddi dysfunction.

PATHOPHYSIOLOGY

- The inciting events in acute pancreatitis begin in pancreatic acinar cells after a primary injury promotes pancreatic enzyme activation (primarily trypsin, although other proteases such as elastase and chymotrypsin may be involved), with subsequent enzymatic “spilling” (Figure 1)
- The enzymes diffuse into the interstitial and endothelial spaces and begin auto-digestion of the gland. Tissue breakdown products potentiate vascular injury, with local recruitment of cytokine and Arachidonic acid metabolite-secreting leukocytes. These agents produce edema and oxidative stress.
- The increase in vascular permeability promotes thrombosis and hemorrhage and can lead to pancreatic ischemia and necrosis. Increased vascular permeability can also lead to bacterial translocation into the pancreatic bed and result in infected pancreatic necrosis, a life-threatening complication of acute pancreatitis.
- In severe cases, systemic inflammatory response syndrome (SIRS), renal failure, shock, myocardial stress, fever, or acute respiratory distress syndrome (ARDS) may develop.

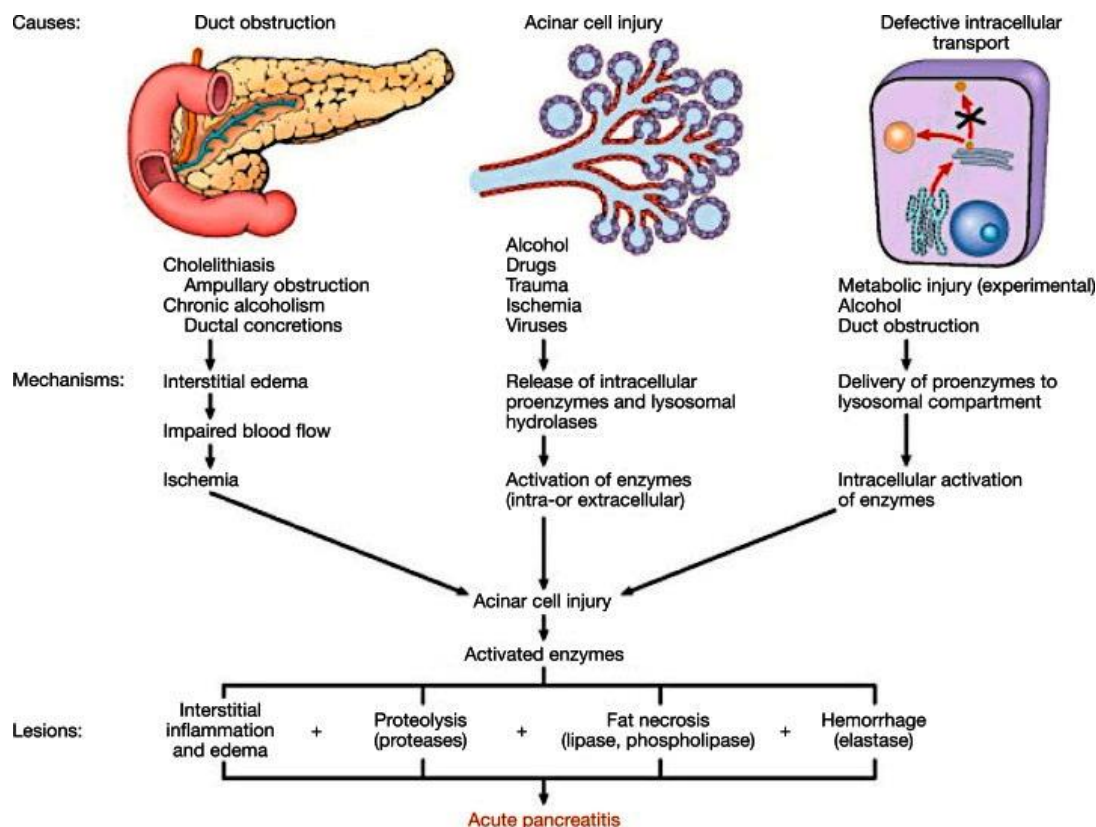


Figure 1: Pathophysiology of acute pancreatitis

- Alcoholic acute pancreatitis may have a slightly different pathogenesis. Alcohol potently stimulates the release of secretin and cholecystokinin, which are the major contributors to pancreatic secretion. Also, the rising ethanol concentration in acinar cells causes an increase in cytosolic calcium, which is required for vesicular zymogen activation. This relationship between cytosolic calcium and zymogen activation may also help to explain the association between hypercalcemia and acute pancreatitis.

DIAGNOSIS

Clinical Presentation

- The classic presentation of acute pancreatitis includes mild to severe epigastric abdominal pain (often radiating to the back) as well as nausea and vomiting. The pain is typically constant in nature and is not aggravated by coughing, movement, or respiration. Pain tends to be more severe in a supine position and may lessen if the patient leans forward in a sitting position.
- Patients may be pale, distressed, have distention, jaundice, tachycardia and fever.
- *Turner's sign* (flank bruising) or *Cullen's sign* (bruising surrounding the umbilicus) may be present in severe cases.
- Some patients may have a more florid presentation that includes hypotension or shock due to intra-vascular volume depletion and third-spacing of fluids.

- Other clinical findings that can be present in acute pancreatitis include dehydration and decreased urine output.
- Findings that may be seen in more severe presentations include hypotension despite volume replacement and a corresponding rise in hematocrit secondary to hemoconcentration, metabolic acidosis, ARDS/respiratory failure, renal failure, and fluctuation in serum calcium levels.

Diagnostic Criteria

- The *Revised Atlanta Criteria* of 2012 (updated from 1992) requires two of three conditions be met to diagnose acute pancreatitis:
 1. Abdominal pain consistent with acute pancreatitis (i.e. epigastric abdominal pain with possible radiation to the back).
 2. Lipase or amylase ≥ 3 times the upper limit of normal, and/or
 3. Characteristic imaging features of acute pancreatitis by CT, MRI, or US.

Biochemical Diagnostic Parameters

- Elevation of serum amylase and lipase levels to >3 times the upper limit of normal in conjunction with the appropriate clinical history are the mainstays in the diagnosis of acute pancreatitis.
- In general, amylase and lipase levels do *not* correlate with either the severity of the attack or with overall prognosis. In addition, serum amylase and lipase levels neither assist in generating an overall prognosis nor in predicting complications of acute pancreatitis.
- A fall in enzymes accompanied by clinical improvement is often an adequate indication that the pancreatitis is resolving in most patients. Persistent elevation of serum amylase and lipase levels may suggest pancreatic ductal disruption and/or necrosis.

Imaging

- **Plain X-ray**

The presence of calcifications may suggest chronic pancreatitis. Signs that may be seen on radiograph in acute pancreatitis include a “sentinel loop,” which represents a dilated segment of small intestine or colon displaying ileus, and “colon cut-off sign,” which represents a functional spasm in the descending colon resulting in a termination of air in the distal colon near the splenic flexure.

- **Ultrasonography (US)**

A hyper-echoic, diffusely enlarged pancreas is often seen on trans-abdominal US in acute pancreatitis. Ultrasonography is also a useful and economic choice for evaluating patients with suspected gallstone pancreatitis. This study can identify stones in the GB, evidence of acute cholecystitis (GB wall thickening or peri-cholecystic fluid), and CBD dilatation (often suggestive of an obstructing CBD stone) and in some cases can directly visualize choledocholithiasis. Bowel gas may obscure the pancreas on US imaging. Ultrasonography does *not* assist in diagnosing the extent of pancreatic necrosis or inflammation.

- **CT scan**

Specific Imaging of the pancreas is recommended only in patients whom the diagnosis is unclear, for those who fail to improve within the first 48–72 hours, or to assess for complications. CT scans with intravenous (IV) contrast are helpful in patients with known or suspected moderate to severe pancreatitis. With this study, the entire pancreas can be well visualized and complications of pancreatitis such as fluid collections, pseudocysts (Figure 2A), and/or areas of necrosis (Figure 2B) can be rapidly identified. There is an excellent correlation between contrast-enhanced CT results and the development of early and late necrosis. The degree of necrosis is also an excellent prognostic factor. However, small areas of necrosis can be missed via contrast CT.

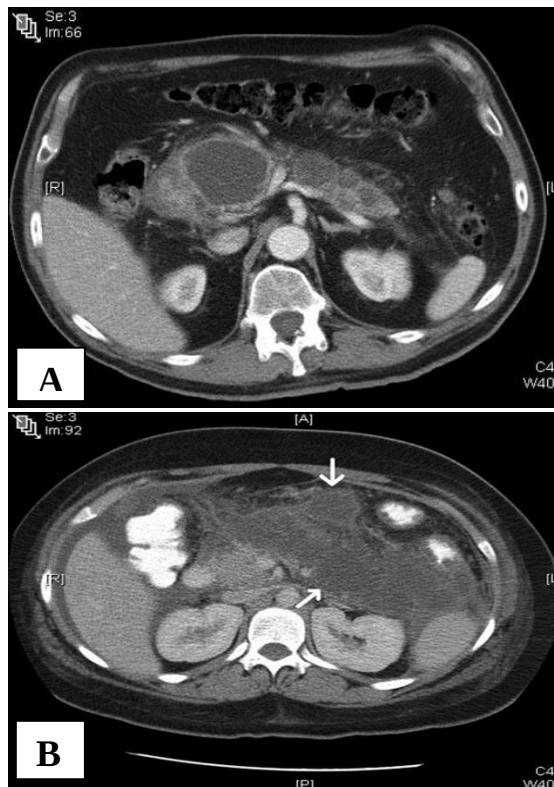


Figure 2. A: Computed tomography (CT) scan obtained from a patient with acute pancreatitis showing multiple well-demarcated pseudo-cysts.

Figure 2B: Computed tomography (CT) scan showing severe necrotizing pancreatitis (arrows).

- **Magnetic resonance imaging (MRI)**

Abdominal MRI is another commonly utilized imaging modality in patients with acute pancreatitis. Studies performed with MRI use *gadolinium* for contrast, which carries a lower risk of side effects or renal injury than contrast used with CT scans. MRI is also highly effective at identifying fluid collections and pancreatic necrosis. It has a greater sensitivity for detecting mild acute pancreatitis as compared with CT scan. MRI may be preferred over CT scan if biliary pancreatitis is suspected as *magnetic resonance cholangio-pancreatography* (MRCP) can be performed.

- **Endoscopic ultrasound (EUS)**

Endoscopic ultrasound can be used to evaluate the CBD for the presence of stones that may require removal via ERCP. If stones are seen via EUS, ERCP can typically be performed at the same time

PROGNOSTIC FACTORS

- Several scoring systems to assess the severity of acute pancreatitis and prognosis of patients (e.g. Ranson's criteria, Glasgow and Acute Physiology and Chronic Health Evaluation II [APACHE-II]) have been developed.
- In 1976, **Ranson** reported the use of a series of 11 objective findings that correlate with severity in patients with acute pancreatitis (Table 2).

Table 2: Ranson's Criteria for Scoring of Acute Pancreatitis

Criteria for Acute Pancreatitis not due to Gallstones	
<i>At Admission</i> - Age >55 y - WBCs >16,000/mm³ - Blood glucose >200 mg/DL - Serum LDH >350 IU/L - Serum AST >250 IU/dL	<i>During the Initial 48 hours</i> - Hematocrit fall > 10 points - BUN elevation >5mg/dL - Serum calcium <8mg/dL - Arterial PO₂ <60 mmHg - Basic deficit >4 mEq/L - Estimated fluid sequestration >6L
Criteria for Acute Gallstone Pancreatitis	
<i>At Admission</i> - Age >70 y - WBCs >18,000/mm³ - Blood glucose >220 mg/DL - Serum LDH >400 IU/L - Serum AST >250 IU/dL	<i>During the Initial 48 hours</i> - Hematocrit fall >10 points - BUN elevation >2 mg/dL - Serum calcium <8 mg/dL - Basic deficit >5 mEq/L - Estimated fluid sequestration >4L

- These scoring systems may be helpful early in the clinical course of acute pancreatitis, although their usefulness is diminished as the disease progresses.
- *Poor prognostic factors:* These include:
 1. Renal failure
 2. Respiratory failure
 3. Multi-organ system failure (MSOF)
 4. Fluid collections
 5. Necrosis
 6. Increased ICU length of stay
 7. Shock.
- *Other prognostic factors* include CT severity index (CTSI) score described by *Balthazar et al* (Table 3).

Table 3: CT Severity Index (CTSI) Score

Grade	Balthazar Score	Points
A	Normal pancreas	0
B	Pancreatic enlargement	1
C	Pancreatic inflammation and/or peripancreatic fat	2
D	Single peripancreatic fluid collection	3
E	Two or more fluid collections and/or retroperitoneal air	4
	Percentage necrosis	
	0	0
	< 30	2
	30 – 50	4
	> 50	6
	CT Severity Index	
	Low degree	0 – 3
	Middle degree (6% mortality)	4 – 6
	High degree (17% mortality)	7 – 10

TREATMENT

- Most patients with *mild acute pancreatitis* recover with supportive measures. Patients should have nothing by mouth status for at least 24-48 hours. In the absence of cardio-pulmonary complications, vigorous hydration with IV fluids should be administered until adequate urine output (UOP) is achieved and maintained. Pain management, typically with narcotics, should be implemented as well. Some data suggest the superiority of a patient-controlled analgesia pump with the agent Meperidine instead of morphine as it might increase sphincter of Oddi pressure. If the underlying cause of the episode of acute pancreatitis is amenable to correction (e.g. choledocholithiasis), therapeutic interventions such as ERCP with biliary sphincterotomy and duct clearance and/or cholecystectomy may be indicated.
- Other important etiologies that may be correctable are alcohol use/abuse, hypercalcemia, hyper-triglyceridemia, and drug-induced pancreatitis.
- There is little to be gained from daily monitoring of serum amylase and lipase levels. Patients can be gradually returned to oral intake as abdominal pain recedes and hunger returns. Over the course of mild acute pancreatitis, most laboratory abnormalities should show improvement and resolution without further intervention within 3-7 days.
- Patients with *more severe acute pancreatitis*, manifested as the development of peripancreatic fluid collection, pancreatic pseudo-cysts, pancreatic necrosis, and/or the development of respiratory, renal, or circulatory compromise, require more aggressive management, and ICU admission is often warranted.
- In general, a multi-disciplinary approach with both medical and surgical teams is generally beneficial. Renal failure may warrant hemodialysis, and patients with respiratory failure may require mechanical ventilation.

- As with mild pancreatitis, nothing by mouth status is recommended in severe acute pancreatitis for at least the first 48 hours after diagnosis. If pancreatic necrosis is seen, enteral feeding is often performed to reduce the risk of bacterial translocation from the gut to the necrotic pancreatic bed, improve intestinal wall integrity, and promote gut motility.
- Parenteral nutrition was once recommended in patients with severe pancreatitis, but it has been associated with increased length of stay, costs, and complication and mortality rates as well as increased systemic and local infections as compared with enteral nutrition. Aggressive pain management and intravenous fluid replacement are recommended. Vital signs and urine output should be monitored every few hours for the first 24 to 48 hours.
- The use of prophylactic antibiotics in patients with pancreatic necrosis (in an attempt to avoid infection) remains controversial. Some studies have demonstrated benefit in this regard, while others have *not* shown an advantage. Imipenem, Meropenem, and Fluoroquinolones are commonly used in this setting as these agents have a high degree of pancreatic penetrance. Although antibiotic prophylaxis cannot be universally recommended in the setting of acute pancreatitis, 4 situations in which administering antibiotics may be appropriate are: systemic inflammatory response syndrome or sepsis, MSOF, proven extra-pancreatic or pancreatic infections, or an increase in C-reactive protein (CRP) with evidence of pancreatic or extra-pancreatic infection.
- Patients who develop *pancreatic necrosis* must be closely monitored for the development of infected pancreatic necrosis. Infected pancreatic necrosis is most commonly treated with surgical debridement, although endoscopic debridement is possible, and percutaneous drainage of pus can be a consideration as well. The timing of surgical intervention varies; surgical therapy may be required urgently if patients are septic. Delaying surgery, if possible, may allow necrotic pancreatic tissue to be well demarcated at the time of surgery. Infected pancreatic necrosis, pancreatic abscesses, and infected pseudocysts are the most common indicators for surgery in the acute phase of the illness, with more minimally invasive techniques favored if possible, although open abdominal procedures may be required.
- Different types of peri-pancreatic fluid collections resulting from an attack of acute pancreatitis are illustrated in Table 4. Most acute fluid collections and/or pseudo-cysts do not require interventions unless they become infected or cause significant extrinsic compression of other organs. In this case drainage is indicated and it is initially attempted via endoscopic/ultrasound-guided maneuvers. Failure to improve after these measures may warrant surgical drainage.

Table 4: Different types of peri-pancreatic fluid collections resulting from an attack of acute pancreatitis

Acute Peripancreatic Collection	Acute Necrotic Collection
- < 4 weeks - In interstitial pancreatitis - Homogeneous - fluid density - <i>No fully definable wall</i> - Adjacent to pancreas - Confined by normal fascial planes	- < 4 weeks - In necrotizing pancreatitis - Heterogeneous collection - <i>No fully definable wall</i> - Intra- or extrapancreatic
Pseudocyst	Walled-off Necrosis
- > 4 weeks - In interstitial pancreatitis - Homogeneous - fluid density - <i>Well defined wall</i> - Adjacent to pancreas - No non-liquid component	- > 4 weeks - In necrotizing pancreatitis - Heterogeneous collection - <i>Well-defined wall</i> - Intra- or extrapancreatic

- General scheme for treatment of acute pancreatitis is outlined in Figure 3.

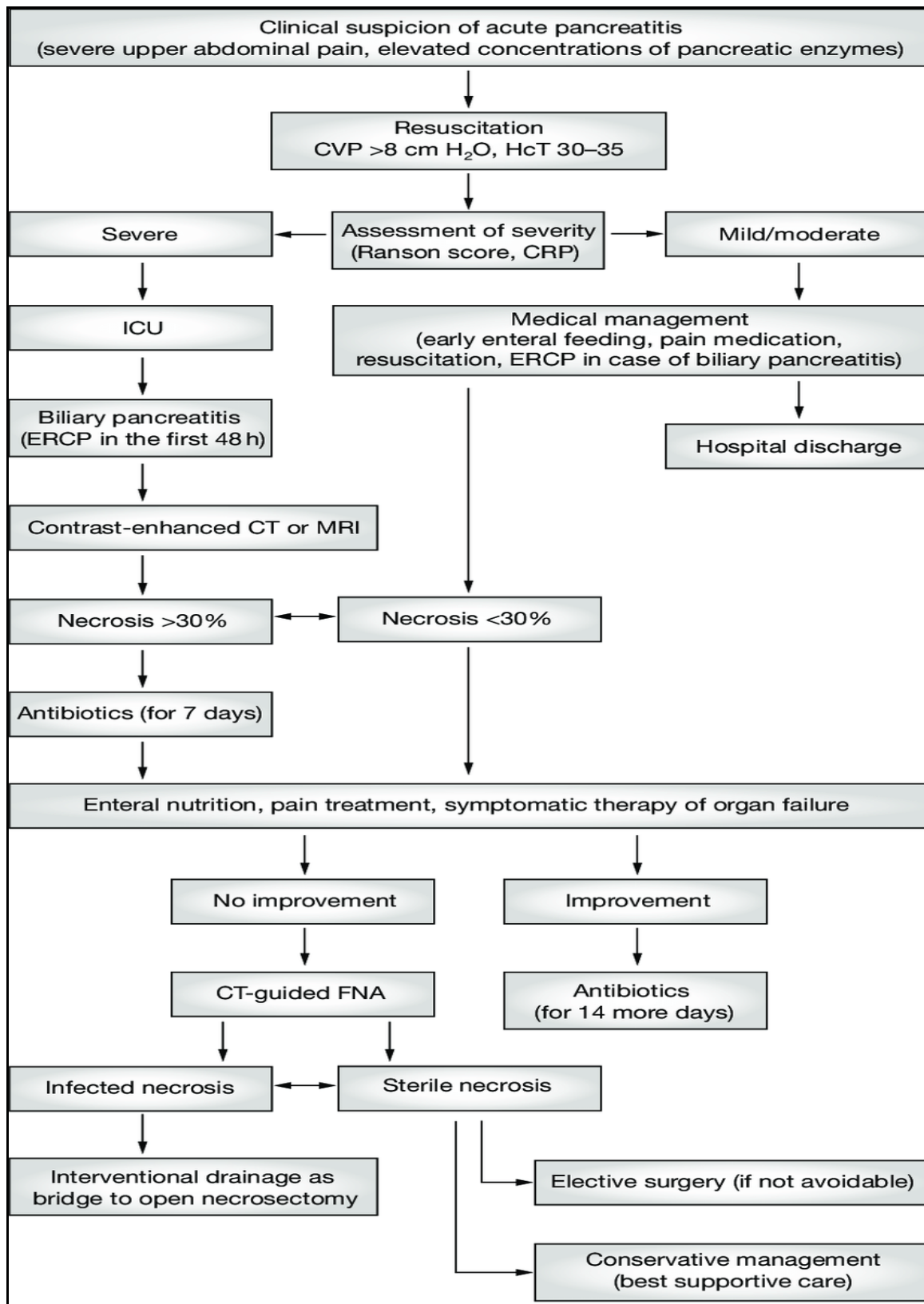


Figure 3: General scheme for treatment of acute pancreatitis

Pancreatic Tumors

CLASSIFICATION OF PANCREATIC TUMORS

According to Cell of Origin

Pancreatic tumors are classified according to the cell of origin into exocrine tumors and endocrine tumors (Table 1).

- **Exocrine tumors:** originating from cells in the exocrine pancreas (ductal and acinar)
- **Endocrine tumors:** originating from endocrinal / hormone-producing cells i.e. A cells, B cells, D cells and D1 cells
- **Benign tumors** of the pancreas are extremely rare

Table 1: Types (Classification) of Pancreatic Neoplasms

Pancreatic Neoplasm	Frequency
1. <i>Pancreatic exocrine neoplasms</i>	90%
- Invasive ductal adenocarcinoma	85%
- Intra-ductal papillary mucinous neoplasms (IPMN)	3-5%
- Serous cystic neoplasms	1-2%
- Mucinous cystic neoplasms	1-2%
- Solid pseudo-papillary neoplasm	1-2%
2. <i>Pancreatic endocrine neoplasm (PEN)</i>	3-4%
3. <i>Miscellaneous epithelial neoplasms</i>	1-2%
- Teratoma	
- Lymphoepithelial cyst	

Pancreatic Ductal Adenocarcinoma (PDAC)

Incidence

- It is the most common type of pancreatic tumors (85-90%)

Risk Factors

1. Cigarette smoking
2. Heavy alcohol consumption
3. Heavy consumption of red or processed meat
4. Chronic pancreatitis
5. Obesity
6. Diabetes Mellites (DM)
 - New onset NIDDM in elderly patients: Older patients >50 years of age with new-onset DM have about an 8-fold elevated risk of having pancreatic cancer compared with the general population
 - Long-term DM (2-8 years duration)

Pre-malignant Pancreatic Lesions

1. Mucinous cystic neoplasms (MCN)
2. Intra-ductal papillary mucinous neoplasms (IPMN)

Genetic Predisposition

- Familial syndromes associated with increased risk of PDAC
- Approximately, 10% of PDAC occur in conjunction with a defined genetic syndrome; these include the following:
 - a) Peutz-Jeghers syndrome.
 - b) Lynch syndrome.
 - c) Familial pancreatitis.
 - d) Familial pancreatic cancer.
 - e) Melanoma-pancreatic cancer syndrome.
 - f) Hereditary breast-ovarian cancer syndrome.

The Peri-ampullary Region

- *Peri-ampullary region* is “the region located within 2 cm around the ampulla of Vater.
- *Peri-ampullary carcinoma* is a term to define carcinomas arising from the head of the pancreas, the ampulla of Vater, the distal common bile duct (CBD) and the duodenum adjacent to the ampulla (Table 2).
- Although, the mode of presentation and treatment options for peri-ampullary tumors are similar, their prognosis are quite different with that for adenocarcinoma of the pancreas being much worse than for the other tumors.
- Most patients with peri-ampullary cancers will present with biliary obstruction due to the location of the tumor. Often, this obstruction will lead to symptoms of obstructive jaundice i.e. dark color urine, clay color stools, and pruritis.

Table 2: Types of Peri-ampullary Carcinoma

Type of Carcinoma	Frequency
1. Adenocarcinoma of head of pancreas	50 %
2. Tumor from the ampulla of Vater	30 %
3. Distal common bile duct carcinoma	10 %
4. Duodenal carcinoma adjacent to ampulla	10 %

Clinical Presentation

- Early pancreatic cancer symptoms are generally non-specific, which contributes to its delayed diagnosis (diagnosis is often delayed until disease is at an advanced stage)
- Only 15- 20% of patients present with *resectable* disease
- Most lesions are *non-resectable* at presentation; the disease is locally advanced in about 25-30% of patients and metastatic in about 50- 60% of patients
- Clinical presentation depends on tumor *location* and *size* (Table 3).
- Location:
 1. Head: 65%
 2. Body and tail: 30%
 3. Diffuse pancreatic involvement: 5%
- Early pancreatic cancer may present with:
 1. Non-specific GIT symptoms such as nausea and vomiting
 2. Mild epigastric pain radiating to the middle of the back
 3. Anorexia and unexplained weight loss.
- Enlarging tumors in the *head of the pancreas* usually present with obstructive jaundice, dark urine, clay stools and pruritis.
- Physical Examination
 1. Jaundice
 2. Epigastric fullness
 3. *Courvoisier* gallbladder (palpable GB)
 4. Scratch marks from pruritus
 5. Migratory thrombophlebitis (Trousseau sign)
 6. Late cases:
 - o Epigastric mass
 - o Ascites
 - o *Troisier sign* (palpable enlarged left supraclavicular lymph node, Virchow node)

Table 3: Differences in presentation between pancreatic head *versus* body and tail lesions

Clinical Presentation	Pancreatic Head	Pancreatic Body and Tail
1) Jaundice	Yes	No
2) Size at discovery	Usually smaller	Usually larger
3) Metastasis at discovery	Less frequent	More frequent

Laboratory Investigations

- Routine lab tests
- Dedicated lab tests
 1. Bilirubin (total and direct)
 2. Markers of cholestasis: alkaline phosphatase (AP), Gamma glutamyl transferase (GGT)
 3. Tumor markers
 - a) Carboxylated Antigen 19-9 (CA19-9):
 - It is *not* tumor-specific i.e. is commonly elevated in pancreatic and hepatobiliary diseases and in many other malignancies.
 - It is high in cases of biliary obstruction regardless of etiology and does not necessarily indicate cancer in jaundiced patients. Pre-operative measurement of CA19-9 is therefore best performed after biliary decompression is complete and bilirubin is normal.
 - Serial measurement of CA19-9 is one of the validated follow-up tools
 - b) Carcino-Embryonic Antigen (CEA)
 - c) CA 125

Imaging Studies: Assessment of Jaundice in Pancreatic Head Lesions

1. *Abdominal Ultrasonography (US)*: It provides sure evidence of biliary tree obstruction namely intra-hepatic bile duct dilatation.
2. *Endoscopic retrograde cholangio-pancreatography (ERCP)*
 - *Diagnostic*: double duct sign or distal CBD stricture
 - *Biopsy*: Punch biopsy from peri-ampullary lesions
3. *Magnetic resonance cholangio-pancreatography (MRCP)*

Imaging Studies: Assessment of the Lesion

1. *Multi-detector computed tomography (MDCT): pancreatic protocol*
 - It is the gold standard for diagnosis and staging.
 - It allows evaluation of resectability through assessing the relation of the tumor to nearby arterial structures and venous structures; namely, the Portal vein (PV), the superior mesenteric vein (SMV), the superior mesenteric artery (SMA) and the celiac artery (CA).
 - It allows staging by detecting the presence or absence of liver metastases and/or lymph nodal spread outside the field of resection.
2. *Magnetic Resonance Imaging (MRI)*:
 - It is most commonly used as a problem-solving tool. When contrast enhanced CT cannot be obtained (severe allergy to iodinated intravenous contrast material)
 - Disadvantages: Higher cost- lack of widespread availability compared to CT
3. *Endoscopic Ultrasound (EUS)*
 - EUS is not recommended as a routine staging study.
 - EUS may be used as a complementary to CT staging
 - Disadvantage: Highly operator-dependent and high cost.

4. Role of Biopsy

- For operable pancreatic masses, biopsy proof of malignancy is **not** required before surgical resection *and a non-diagnostic biopsy should not delay surgical resection when the clinical suspicion of pancreatic cancer is high*
- Pancreatic biopsy should be reserved for patients with locally unresectable or metastatic disease prior to chemotherapy.
- Biopsy of a pancreatic lesion should be performed endoscopically with either ERCP or EUS-FNA.
- Percutaneous biopsy of pancreatic mass is generally inadvisable owing to potential for tumor seeding along the needle track. Percutaneous US- or CT-guided pancreatic biopsy should be reserved for evaluation of suspected metastatic lesions, typically liver metastases and *not* for the lesion itself.

5. Positron Emission Tomography (PET) Scan

- It is sensitive in detecting extra-pancreatic metastatic disease of questionable morphology.

6. Diagnostic Staging Laparoscopy:

- It is used in some institutions to rule out peritoneal metastases *not* detected on imaging, particularly those that may occur in body and tail of pancreas.

At the end of assessment, lesions should be classified into either: resectable, border-line resectable, or unresectable (Table 4)

Table 4: Criteria Resectability and Irresectability of Pancreatic Cancer

Resectable	Borderline Resectable	Unresectable
No distant metastases	No distant metastases	Distant metastases
No arterial contact	Less than 180-degree SMA encasement.	Greater than 180-degree SMA encasement
No venous contact	Reconstructable SMV/portal vein contact	Non-reconstructable SMV/portal vein contact

Treatment

1. Resectable Tumors

A. For tumors in the head of the pancreas:

Surgical resection with the intent of cure in the form of pancreatoduodenectomy (Whipple procedure) in which the followings are resected:

- pancreatic head and neck
- distal 40% of the stomach
- whole duodenum
- proximal 10% of the jejunum
- lower part of the common bile duct (CBD) along with the gallbladder (GB)

Then the continuity of the gastrointestinal tract (GIT) is restored via

- Pancreatico-jejunostomy
- Hepatico-jejunostomy
- Gastro-jejunostomy

B. For tumors in the body and tail of the pancreas:

- Surgical resection in the form of distal pancreatectomy and splenectomy

2. *Border Line Resectable Tumors*

- *Neoadjuvant chemotherapy* is given pre-operatively to potentially permit cases with borderline resectability to achieve a partial response with therapy in the hope of improving chance for a complete, margin-free (R0) resection.
- This is followed by reassessment by MDCT, if it becomes resectable, trial resection is attempted. This decision is made under the direction of multidisciplinary surgeon and oncologist consultations

3. *Unresectable Tumors: Palliative measures for*

a) Biliary obstruction:

- Endoscopic stenting
- Percutaneous trans-hepatic biliary drainage
- Surgical bilio-enteric bypass i.e. hepatico- or cholecysto-jejunostomy

b) Gastric outlet obstruction:

- Surgical gastro-jejunostomy
- Endoscopic stenting

c) Severe tumor-associated pain:

- Narcotics administration
- Celiac plexus neurolysis

PANCREATIC ENDOCRINE NEOPLASMS

The common pancreatic neuro-endocrine tumors (pNETs) are listed in Table 5, which summarizes the hormone causing the syndrome, signs and symptoms as well as the frequency of the tumor or syndrome. Table 6 summarizes the malignant potential of pancreatic endocrine tumors.

Table 5: Summary of Common Pancreatic Neuro-Endocrine Tumors (pNETs)

Name of Tumor (Syndrome)	Hormone Causing the Syndrome	Signs or Symptoms	% of All functional pNETs
Insulinoma	Insulin	Hypoglycemia (weakness, sweating, tremors, palpitation, confusion, visual changes, etc)	35-40 %
Gastrinoma (Zollinger-Ellison)	Gastrin	Abdominal pain, refractory peptic ulcer disease, secretory diarrhea	16-30 %
Glucagonoma	Glucagon	Dermatitis (migratory necrolytic erythema, DM, diarrhea, DVT (4D-syndrome)	<10 %
VIPoma (Verner-Morrison)	VIP	Profuse watery diarrhea, severe dehydration, hypokalemia, achlorhydria (WDHA syndrome)	<10 %
Somatostatinoma	Somatostatin	DM, cholelithiasis, steatorrhea, anemia, steatorrhea	<5 %

pNETs: Pancreatic Neuro-Endocrine Tumors: DM: diabetes mellitus, VIP: vasoactive intestinal peptide, DVT: deep vein thrombosis

Table 6: malignant potential of pancreatic neuro-endocrine tumors (pNETs)

Type of Pancreatic Neuro-Endocrine Tumors	Malignant Potential
1. Functioning (hormone-producing)	
- Insulinoma (B cells)	10 %
- Gastrinoma	60 %
- Glucagonoma (A cells)	80 %
- Somatostatinoma (D cells)	90 %
- VIPoma	80 %
- PPoma (PP cells)	80 %
2. Non-functioning Tumors	92 %

Liver Cysts and Abscesses

OVERVIEW

- The precise prevalence and incidence of liver cysts are not known because most do not cause symptoms; however, liver cysts have been estimated to occur in 5% of the population. No more than 10-15% of these patients have symptoms that bring the cyst to clinical attention.
- Cystic lesions of the liver include the following:
 1. Simple cysts
 2. Multiple cysts arising in the setting of polycystic liver disease (PCLD)
 3. Parasitic or hydatid (Echinococcal) cysts
 4. Cystic tumors
 5. Abscesses
 6. Ductal cysts (choledochal cysts and Caroli's disease) are differentiated from hepatic cysts by involvement of the bile ducts.
- The evaluation of a patient with a simple liver cyst involves carefully recording the patient history and performing a physical examination plus an imaging study to define the anatomy of the cyst.
- The imaging in patients with hepatic cysts usually includes ultrasonography (US), which is readily available, non-invasive, and highly sensitive and/or computed tomography (CT) scan, which is also highly sensitive and is easier for most clinicians to interpret, particularly for treatment planning.
- Other tests are generally *not* necessary in the evaluation of hepatic cysts. Percutaneous aspiration should be avoided because the laboratory and cytological evaluation of the simple cyst fluid is non-diagnostic, and a small risk exists of inducing anaphylaxis from leakage from the hydatid cyst or of causing abscess formation in a previously sterile cyst.
- Histological assessment of the excised cyst wall should be routinely undertaken to identify the presence of an unsuspected neoplasm, such as cystadenoma.

SIMPLE LIVER CYSTS

- **Etiology:** The cause of simple liver cysts is not known, but they are believed to be *congenital* in origin.
- **Pathology:** The cysts are lined by biliary-type epithelium. Typically, the fluid within the cyst has an electrolyte composition that mimics plasma. Bile, amylase, and white blood cells (WBCs) are absent. The cyst fluid is continually secreted by the epithelial lining of the cyst. For this reason, needle aspiration of simple cysts is *not* curative, and recurrence is the norm.
- **Clinical presentation:** Simple cysts generally cause no symptoms, but may produce dull right upper-quadrant pain if large in size. Patients with symptomatic simple liver cysts may also report abdominal bloating and early satiety. Occasionally, a cyst is large enough to produce a palpable abdominal mass. Jaundice caused by bile duct obstruction is rare, as are cyst rupture and acute torsion of a mobile cyst.
- **Investigations:** Liver function test (LFT) results, such as transaminases or alkaline phosphatase, may be mildly abnormal, but bilirubin, prothrombin time (PT), and activated partial thromboplastin time (aPTT) are usually within normal range. Simple cysts have a typical radiographical appearance. They are thin-walled and uniform with a homogenous low-density interior on CT scan.
- **Management:** Treatment of simple cysts of the liver is indicated only in symptomatic patients. Asymptomatic patients do not require therapy, because the risk of developing complications related to the lesion is lower than the risk associated with treatment. When the cysts become large and cause symptoms, such as pain, treatment is warranted. Surgical treatment of simple liver cysts involves "deroofting" the cyst by excising the portion of the wall that extends to the surface of the liver. Excision of this portion of the cyst wall at the liver surface produces a saucer-type appearance in the remaining cyst so that any fluid secreted from the remaining epithelium leaks into the peritoneal cavity where it can be absorbed. The cyst wall should be sent to pathology to confirm the diagnosis and exclude cystadenoma or cystadenocarcinoma. In simple cysts, histology of the cyst wall generally reveals a layer of simple cuboidal epithelium. Historically, treatment of symptomatic hepatic cysts required laparotomy, but today, cyst deroofing can be successfully performed laparoscopically and is currently considered the standard of care.

POLYCYSTIC LIVER DISEASE (PCLD)

- **Etiology:** Adult PCLD (AD-PCLD) is *congenital* and is usually associated with *autosomal dominant polycystic kidney disease* (AD-PKD). In patients with PKD, the kidney cysts usually precede the liver cysts. PKD often results in renal failure, whereas liver cysts only rarely are associated with hepatic fibrosis and liver failure.
- **Clinical Presentation:** Polycystic liver disease (PCLD) rarely arises in childhood. These cysts are observed at the time of puberty and increase in adulthood. Women are more commonly affected, and an increase in cyst size and number is correlated with estrogen

level. In PCLD, hepatomegaly may be prominent, and patients occasionally progress to hepatic fibrosis, portal hypertension, and liver failure. *Complications* (e.g. rupture, hemorrhage, and infection) are rare. However, patients do present with abdominal pain as the cysts enlarge.

- **Investigations:** In the setting of PCLD, greater abnormalities in LFT results are found, but liver failure is rare. Renal function test results, including blood urea nitrogen (BUN) and creatinine levels are often abnormal and should be performed on initial evaluation. Diagnosis of PCLD is confirmed by means of US or CT scan with multiple liver cysts identified.
- **Management:** Treatment of PCLD is indicated only in symptomatic patients. Only those patients with clearly disabling pain should be considered for surgery. The surgical goal is to decompress as much of the cystic liver as possible. This can be accomplished by a combination of deroofing and fenestration or, in selected patients, by resection of the involved portion of the liver. Recurrence of symptoms with either procedure is high as new cysts replace those that have been resected. Small numbers of patients have been treated with liver transplantation.

NEOPLASTIC CYSTS

- **Etiology:** Liver tumors with central necrosis visualized on imaging studies are often misdiagnosed as liver cysts. True intra-hepatic neoplastic cysts are rare. The cause of cyst-adenomas and cyst-adenocarcinomas is unknown.
- **Pathology:** These cystic tumors are lined with biliary-type cuboidal or columnar cells and are surrounded by ovarian-like stroma. Cystadenoma is a pre-malignant lesion with neoplastic transformation to cyst-adenocarcinoma confirmed by tubulo-papillary architecture and invasion of the basement membrane.
- **Clinical Presentation:** Cystadenoma most often occurs in middle-aged women. However, cyst-adenocarcinoma equally affects both, men and women. Most patients are asymptomatic or have vague abdominal complaints of bloating, nausea, and fullness. These patients, like all those with hepatic cysts, eventually present with abdominal pain. Rarely, they present with evidence of biliary obstruction.
- **Investigations:** With cystic tumors, as with simple cysts, LFT results are normal. There may be mild abnormalities in some patients. Carbohydrate antigen (CA) 19-9 levels are elevated in some patients. Cystadenoma and cyst-adenocarcinoma usually appear multi-loculated with internal septations, heterogeneous density, and thicker, irregular hypervascular cyst wall. Unlike many tumors, cystadenoma and cystadenocarcinoma are rarely associated with calcifications.

HYDATID CYSTS

- **Etiology:** Hydatid cysts are caused by infestation with the parasite *Echinococcus granulosus*. This parasite is found worldwide, but it is particularly common in areas of sheep and cattle farming.

- **Pathogenesis / Pathology:** The adult tapeworm lives in the digestive tract of carnivores, such as dogs or wolves. Eggs are released into the stool and are inadvertently ingested by the *intermediate hosts*, such as sheep, cattle, or humans. The egg larvae invade the bowel wall and mesenteric vessels of the intermediate host, allowing circulation to the liver. In the liver, the larvae grow and become encysted. The hydatid cyst has 3 layers: (1) *adventitia*, an inseparable fibrous tissue due to reaction of the liver to the parasite (2), *laminated membrane (ectocyst)*, formed of the parasite itself; it is whitish, elastic, which can be peeled-off readily from the adventitia, and (3) *germinal epithelium (endocyst)*, the only living part, lining the cyst and produces daughter cysts. When carnivores ingest the liver of the intermediate host, the scolices of the daughter cysts are released in the small intestines and grow into adult worms, thus completing the life cycle of the worm.
- **Clinical Presentation:** Patients with hydatid cysts, like those with simple cysts, are most often asymptomatic, but pain may develop as the cyst grows. Larger lesions typically cause pain and are more likely to develop complications than simple cysts. At presentation, patients generally have a palpable mass in the right upper quadrant.
- **Complications:** Cyst rupture is the most serious complication of hydatid cysts. Cysts may rupture into the biliary tree, through the diaphragm into the chest, or freely into the peritoneal cavity. Rupture into the biliary tree may result in jaundice or cholangitis. Free rupture into the peritoneal cavity may cause anaphylactic shock. As with simple cysts, patients with hydatid cysts may develop secondary infection and subsequent hepatic abscesses.
- **Investigations:** Eosinophilia is noted in approximately 40% of patients, and Echinococcal antibody titers are positive in nearly 80% of patients. *Casoni's test* (intradermal test—70% sensitive—historical interest. Hydatid cysts can be identified by the presence of daughter cysts within a thick-walled main cavity. Ultrasound may reveal rosettes of daughter cysts, double-contoured membrane of the cyst, and calcification of cyst wall (in adventitia). CT scan abdomen is more accurate in identifying cyst characteristics — *cart-wheel-like*—multi-vesicular rosette-like, calcification, floating membrane, and hydatid sand. In patients who are jaundiced with hydatid disease, MRCP or ERCP should be performed to determine whether the cyst has ruptured into the bile duct.
- **Management:** Patients with hydatid cysts should be treated to prevent complications related to cyst growth and rupture.
 1. *Medical therapy* with anti-hydatid agents (Albendazole and Mebendazole) is relatively ineffective. These drugs are used as adjuvant treatment, but they do not replace surgical or percutaneous therapy in symptomatic patients. However, they may have a role in small accidentally discovered hydatid cysts. In surgically treated patients, the use of anti-hydatid agents is generally given peri-operatively; continuation is limited to those who have spillage of cyst fluid at the time of operation or to those with cyst rupture. Anti-hydatid agents are used in conjunction with percutaneous treatment. Medical therapy should be started 4 days before percutaneous treatment and continued either for 1 month (Albendazole) or for 3 months (Mebendazole).

2. **PAIR** (puncture, aspiration, injection, and re-aspiration) is a percutaneous treatment technique for hydatid disease. In this minimally- invasive method, a needle is introduced into the cyst under US guidance. Cyst fluid is aspirated and analysed. Hypertonic saline is then injected and re-aspirated. Its use in the treatment of hydatid cysts has been somewhat controversial. However, as this technique safety and efficacy have been reported in the literature, it has been increasingly accepted as a treatment option for hydatid disease although its use is still limited. It is contraindicated in non-cooperative patients and in Inaccessible or risky location of the liver cyst and cyst communicating with the biliary tree.
3. Most liver hydatid cysts are managed *surgically*. Treatment of hydatid cysts is associated with two technical problems; first, risk of anaphylaxis from spillage of cyst fluid containing eggs and larvae into the peritoneal cavity and, secondly, recurrence caused by residual eggs in incompletely removed germinal membranes. To prevent these problems, the cyst contents are aspirated and replaced with a hypertonic saline solution to kill residual daughter cysts in the germinal membrane. The abdomen is opened, and the peritoneal cavity is packed with mops [black or colored mops are used to identify white scolices clearly so as to pick up all and prevent any spillage]. Fluid from the cyst is aspirated and scolicidal agents [Cetrimide, Chlorohexidine, alcohol, hypertonic saline (15%-20%), 10% Povidone iodine or H₂O₂] are injected into the cyst cavity (formalin should not be used). The cyst is then opened at most superficial part, contents aspirated. The goal of the procedure is to excise the lamellated and germinal membrane, leaving the fibrous component of the cyst wall in situ (cystectomy). The residual cavity is usually packed by a pedicled omentum. *Laparoscopic cystectomy* is becoming more popular. Contraindications are deeply situated cyst, densely adherent cyst, and inaccessible cysts. Main problem with laparoscopic cystectomy is spillage and difficulty in preventing it. Attempts to excise the entire cyst wall without opening the cyst (peri-cystectomy) or to perform formal hepatectomy for hydatid cysts have largely been abandoned because of increased surgical morbidity. However, Liver resection is occasionally done in peripherally situated cyst or if diagnosis of cystic tumour could not be excluded.

AMEBIC LIVER ABSCESS

- **Incidence:** Nowadays, amebic liver abscess (also known as tropical abscess) is generally less common than pyogenic abscess.
- **Etiology:** *Entameba histolytica* is the causative agent. It is contracted by ingestion of food or water contaminated by the cyst stage of the parasite. Amebiasis generally only involves the intestine but can invade the mesenteric venules resulting in liver abscesses. Its only host is the human.
- **Clinical Presentation:** The patient is *toxic* with fever, chills and rigors, loss of appetite, reduced weight, and jaundice. The abdomen shows localized guarding and rigidity, as well as a mass in right upper abdomen (tender, soft liver). Patients may have a history of diarrhea and dysentery.

- **Sequelae and complications:** An amebic abscess can rupture into the peritoneum, pleural cavity, pericardial cavity, bronchus (causing a broncho-pleural fistula leading to coughing out of anchovy sauce pus), bare-area of liver (causing retro-peritoneal abscess), the intestines, or into the skin (amebiasis cutis).
- **Investigations:** Total WBC count may be increased. Liver function tests may show altered bilirubin and albumin. Serum alkaline phosphatase, AST and ALT levels are elevated. Serological tests (ELISA) are reliable in non-endemic areas. *US abdomen* shows altered echogenicity (an-echogenic, hypo-echogenic), size, location, number of abscess, nature of the liver. *CT scan* shows raised diaphragm; abscess cavity (low density area)—its size, location, number; presence of effusion; changes in the lung. *Sigmoidoscopy/colonoscopy* is used to identify the active ulcers. *Scrapings* of the ulcer show trophozoites.
- **Management:** Treatment is mainly *medical* using metronidazole. *Aspiration* with injection of amebicidal agent is indicated in case of large abscess done under US guidance. Aspirated fluid is sent for culture and sensitivity, cytology and for study of trophozoites.

PYOGENIC LIVER ABSCESS

- **Etiology**
 1. Ascending cholangitis: commonest route.
 2. Portal vein sepsis: appendicitis, diverticulitis, inflammatory bowel disease (IBD), etc.
 3. Distant infections (through hepatic artery)
 4. Super added infections: amoebic liver abscess, hydatid cyst.
 5. Cryptogenic liver abscess — No identified primary infection.
 6. Trauma; becoming a common cause.
 7. Direct extension: from suppurative cholecystitis, subphrenic abscess, perforation, peri-nephric abscess.
 8. Umbilical vein sepsis.
- **Clinical Presentation:** It is of increasing incidence especially in elderly and diabetic patients or immuno-compromised individuals. It may complicate cholangitis, appendicitis, or diverticulitis. Clinical diagnosis is suggested when anorexia, fever, malaise, upper right abdominal discomfort are present.
- **Investigations:** Micro-organisms isolated are most often bowel flora. *E. coli* (commonest), *Klebsiella*, *Proteus*. Enterococci, streptococci viridians in polymicrobial infection. Leukocytosis is generally present. Abdominal US and CT scan are diagnostic. US-guided aspiration of pus for culture and sensitivity is usually needed. Blood culture may be positive.
- **Management:** Patients with liver abscesses are immediately started on antibiotics according to culture and sensitivity. If abscesses are small, patients may respond to *medications* alone. More likely, these patients will require the addition of *percutaneous drainage*. Percutaneous drainage and antibiotics are usually adequate treatment. If abscesses persist despite attempted percutaneous drainage, *surgical drainage* is

indicated. Other surgical indications include large cysts at risk of rupture and abscesses not anatomically amenable to percutaneous treatment. Treating the primary causes should not be forgotten. In general, abscesses are adequately managed by means of antibiotics and percutaneous drainage.

Liver Tumors

Classification (Table 1)

Table 1: Benign and malignant tumors of the liver

Benign	
- True neoplasms	- Liver cell adenoma (LCA)
- Non-neoplastic lesions	- Hemangioma - Focal nodular hyperplasia
Malignant	
- Primary hepatocytes	- Hepatocellular carcinoma (HCC) (hepatoma)
- Bile ducts	- Cholangio-carcinoma
- Mixed	- Cholangio-hepatoma
- Secondary	- From GIT, pancreas or breast cancer

BENIGN TUMORS OF THE LIVER

Hemangioma

- *Incidence:* It is the *commonest* BT of the liver. It affects all ages with a mean age in adults 50 years AND occurs in females > males (5:1). It affects both lobes equally.
- *Etiology is not well-understood*, but may be related to estrogen receptors (accelerated growth with high estrogen states; puberty, pregnancy and oral contraceptive pills-OCPs).
- *Pathology*
 1. *Grossly:* Well-delineated, flat, red-blue lesions that may coalesce partially on sectioning ± fibrosis, calcification and thrombosis. It may attain a large size of 10 cm or more (*giant hemangioma*) (Figure 1).
 2. *Microscopically:* It is of the *cavernous* type showing cavernous vascular spaces lined by flattened epithelium underlying fibrous septae of various widths.
- *Clinically*
 1. It is usually < 5 cm, harmless and found incidentally during US or CT scan for unrelated reason. It may remain stable in size or increase minimally over time.
 2. Pain related to uncomplicated hemangioma is probably due to an associated disorder.
 3. Large hemangiomas can be asymptomatic or manifest with pain or an abdominal mass.
 4. Large lesions in the left lobe may cause pressure effects.

- *Complications*
 1. Alteration of internal structure, such as inflammation (low-grade fever, weight loss, abdominal pain, ↑ ESR, anemia, thrombocytosis and ↑ fibrinogen level).
 2. Alteration in coagulation, which → systemic disorders (**Kasabach-Merritt** syndrome: intra-vascular coagulation, clotting and fibrinolysis within the hemangioma, which may progress to → 2nd ↑ systemic fibrinolysis and thrombocytopenia → death (20-30%).
 3. Hemorrhage → hemo-peritoneum (internal hemorrhage). It is rare and may result from spontaneous rupture or after anti-coagulation therapy.
- *Diagnosis*
 1. *US* shows a focal lesion, a classically echogenic mass of uniform density, < 3 cm in diameter, with acoustic enhancement and sharp margins.
 2. *CT scan* (Figure 2): Criteria of Dx are (1) low attenuation on non-contrast CT, (2) peripheral enhancement of the lesion followed by central enhancement on contrast CT and (3) contrast enhancement of the lesion on delayed scans.
 3. *Dynamic MRI*: Hemangioma is a hypo-intense lesion on T1-weighted sequences and strongly hyper-intense on T2-weighted sequences with a "light bulb pattern".
 4. *Biopsy* is *contra-indicated* if hemangioma is suspected.



Figure 1. Gross appearance of hemangioma



Figure 2. CT scan showing hemangioma (arrow)

- *Treatment*
 1. *Asymptomatic* hemangiomas that are diagnosed at laparotomy should *not* be resected as the natural history is towards involution and calcification. Just patient assurance is required as it has *no tendency* to turn to malignancy.
 2. *Surgical resection* is indicated in presence of (1) severe symptoms, (2) complications and (3) to exclude malignancy. It includes enucleation or hepatic resection (laparoscopic) depending on the size and anatomical location of the lesion.

Focal Nodular Hyperplasia (FNH)

- It is the 2nd most common BT of the liver that usually affects women between 30-50 years.
- This is a harmless lesion of the liver whose main importance is the difficult differentiation from other focal lesions.
- *Etiology*: Hyperplastic reaction resulting from arterial malformation (↑ arterial flow → hyper-perfusion of local liver parenchyma → 2nd hepatocellular hyperplasia) (a regenerative procedure). Contraceptive pills stimulate the development & growth of FNH.

- *Pathology*

1. *Grossly*, FNH is a nodular firm mass, which is slightly paler than the normal liver and has prominent surface vasculature. The cut-section shows a *stellate scar* with radiating fibrous septa that divide the lesion into lobules (Figure 3).
2. *Histologically*, it resembles cirrhosis with regenerating nodules. The central scar and fibrous septa are apparent



Figure 3. Focal nodular hyperplasia

- *Clinical presentation*

1. Asymptomatic.
2. Abdominal pain or discomfort caused by pressure of large FNH or severe pain due to torsion of pedunculated lesions.

- *Complications:*

1. Rupture
2. Bleeding (rare). No risk of malignancy.

- *Diagnosis / Investigations*

1. *US*: It is hyper- or iso-echoic on US examination. The central scar is slightly hyper-echoic (difficult to visualize). Color Doppler shows a central feeding artery with a stellate or spoke-wheel pattern corresponding to the artery running from the central scar to fibrous septa.
2. *CT scan*: It also shows a hypo- or iso-echoic mass. The central scar is seen in 30% of cases only. FNH enhances at the arterial phase, but the central scar is hypodense. The lesion returns to become iso-dense in the portal phase.
3. *MRI*: A hypo- or iso-intense lesion is seen on T1-weighted images and iso- or slightly hyper-intense on T2- weighted images.

- *Treatment*

1. Asymptomatic lesions require *no* treatment regardless of their number or size. Patient assurance is needed and contraceptive pills should be discontinued.
2. Surgical resection is indicated in symptomatic patients or in doubtful cases i.e. cases that are suspicious of malignancy are explored and subjected to frozen section examination, which is diagnostic.

Liver Cell Adenoma (LCA)

- *Definition*

This lesion is a *rare* true benign neoplasm of hepatocytes, usually solitary (but may be multiple) and sometimes pedunculated.

- *Etiology*

1. It occurs almost only in women. It is strongly associated with OCPs and androgen steroid therapy. Cessation of pill-intake can result in tumor regression.
2. It can occur spontaneously or associated with underlying metabolic disease including Type I glycogen storage disease, iron overload related to β -Thalassemia and DM.

- *Pathology*

1. *Grossly*, it is soft, well-circumscribed and light yellow in color (Figure 4). It is multiple in 1/3 of cases. It is sometimes encapsulated with fleshy appearance $\pm$ areas of hemorrhage and necrosis.
2. *Histologically*, it is formed of sheets of regular hepatocytes with thin-walled vessels, yet no portal triads.

- *Clinical features / complications*

1. Small LCA is *asymptomatic* and is discovered incidentally on US or at laparotomy.
2. Sometimes, a large LCA causes right upper quadrant pain or discomfort.
3. Complications are rare and include malignant transformation (10%) and spontaneous rupture causing internal hemorrhage (20-40%), more in women, during pregnancy.

- *Investigations*

1. *US*: Well-delineated heterogenous liver mass. Hyper-echogenicity is due the presence of fat and glycogen.
2. *CT scan*: Hyper-vascular & heterogenous on the arterial phase and becomes iso- or hypo-dense on the portal phase due to arterio-venous shunting. Distinctive features from FNH are the smooth surface, present of hemorrhage and necrosis & the tumor capsule.
3. *MRI*: It is the *best tool* because of its higher sensitivity for fat and hemorrhage. Adenomas appear iso- or hyper-intense (due to fat content) on T1-weighted images and mildly hyper-intense on T2-weighted images. Heterogeneity of signal intensity has been considered one of the most constant features of hepatocellular adenoma.

- *Treatment*

1. *Liver resection* is indicated if doubt exists as to the possibility of malignancy, and for large (> 4 cm) symptomatic tumors.
2. In other cases (small lesions < 3 cm), the tumor is regularly checked for any change in size (it may regress), contraceptive pill-intake is discontinued and pregnancy avoided.



Figure 4. Well-circumscribed adenoma within the liver (arrow).

MALIGNANT TUMORS OF THE LIVER

A. PRIMARY LIVER CANCER

Pathological Types

- Hepatocellular carcinoma (HCC)
- Hepatoblastoma (in infancy)
- Mixed HCC and cholangiocarcinoma.
- Intra-hepatic cholangiocarcinoma.

Epidemiology

- In contrast to BTs, 1st liver malignancies are, by far, commoner in men.
- Hepatocellular carcinoma (HCC) is the main 1st liver malignancy and is prevalent in the Far East and in equatorial African nations. The world's highest incidence is in *Mozambique*.
- Cholangio-carcinoma arising from the intrahepatic bile ducts is much less common than HCC and it also has its highest incidence in the *Far East*.

Etiology

Hepato-cellular Carcinoma (HCC)

- The following conditions are associated with the development of HCC:
 1. Infection with *hepatitis B virus* (HBV), or *hepatitis C virus* (HCV).
 2. Liver *cirrhosis* of any cause. The highest incidence of HCC is with post-hepatic, followed by alcoholic, cirrhosis (90% of HCC occurs in patients who are carriers of hepatitis virus or who have cirrhosis). Prolonged infection with HBV or HCV → integration of the viral DNA into the host genome, thus starting the malignant changes.
 3. *Aflatoxin ingestion*: This substance is formed by the *fungus Aspergillus flavus*, which grows on grains that are stored under moist warm conditions.

Cholangiocarcinoma

- Development is related to infestation with *Clonorchis sinensis* (liver *fluke*) that causes *Asciatic cholangitis*.

Pathology

Hepato-cellular Carcinoma (HCC)

- *Gross appearance*: The tumor is yellow in color & commonly arises in a cirrhotic liver. It assumes different forms:

1. *Massive form*, i.e. forming a localized mass (Figure 5). The *fibro-lamellar* variant of HCC is a variety of the massive type that affects non-cirrhotic young females & hence has a *better* prognosis.
2. *Nodular form*, i.e. multiple nodules scattered over the liver.
3. *Diffuse form*, characterized by diffuse infiltration throughout the liver.

- *Microscopically*: HCC is formed of malignant hepatocytes with little stroma, but high vascularity from the HA. Cells possess acidophilic cytoplasm and large nuclei. In well-differentiated cases, cells are similar to normal liver cells, but in anaplastic cases the similarity is less striking. The fibro-lamellar type is an exception as it contains numerous fibrous septa that make it resemble FNH & the cells are deeply eosinophilic.

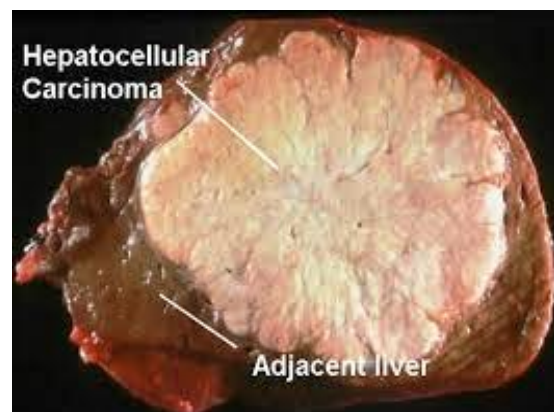


Figure 5. Hepatocellular carcinoma (HCC)

- *Complications*
 1. *Spread* by the lymphatic and venous routes producing porta hepatis nodal enlargement, peritoneal nodules and less commonly lung deposits.
 2. *Spontaneous rupture* → subcapsular hematoma or massive intra-peritoneal hemorrhage.

Intra-hepatic Cholangio-carcinoma

- The tumor is an *adenocarcinoma* of the lining epithelium of the intra-hepatic bile ducts.
- It spreads widely inside and outside the liver, and by the time the tumor is discovered, the disease is usually so advanced to cure.

Hepato-cholangio-carcinoma

- This is a *mixed type* that behaves like HCC.

Hepatoblastoma

- This is a form of HCC that occurs in children. It is termed hepatoblastoma because of the similarity to fetal liver cells.

Other 1st Malignant Tumors of the Liver

- These include: Angiosarcoma, epitheloid hemangio-endothelioma, undifferentiated embryonal sarcoma, and leiomyosarcoma

Clinical Features

Different types of presentation depend on the race and Country + tumor type and size.

1. Deterioration of health of a known cirrhotic is a common presentation.
2. Small lesions may be accidentally discovered by an US done for another purpose.
3. Where HCC is a common, an early detection program is recommended by annual US and AFP estimation (↑ AFP or appearance of a focal lesion on US, warrants further investigations for early detection of the tumor).
4. HCC in black Africans grows rapidly producing pain, jaundice, fever and a tender mass, which is commonly mistaken for an amebic liver abscesses.
5. Late cases present by pain, jaundice, ascites, hepatomegaly, anorexia, loss of weight and probably massive intra-peritoneal hemorrhage.

Investigations

Laboratory Tests

1. Serum AP ↑. The rest of LFTs may show the pattern of cirrhosis if present.
2. Serum AFP is a tumor marker that is present in the serum of the fetus. The normal adult level is 0-10 ng/mL. Its serum level ↑ highly in HCC and testicular teratoma. A level >200 ng/mL is suggestive and >2000 ng/mL is diagnostic of HCC. It may also ↑ in other *benign* liver diseases, but to a lesser extent. Serum AFP returns to normal level after a successful tumor resection. Re-elevation in the follow-up period signifies a recurrence.

Imaging

1. *US Abdomen*: It can detect the focal hepatic lesion, P) patency and presence of ascites.
2. *Tri-phasic CT scan* (Figure 6) *and/or MRI* can detect HCC (number, site, size vascular invasion) and extent of tumor spread in the liver. It can also detect portal vein (PV) invasion and ascites.
3. *Chest X-ray, CT scan and PET scan* to search for chest metastases.

Biopsy

- The value of pre-operative guided-biopsy is *controversial* because of the possible risk of hemorrhage from the highly vascular tumor.

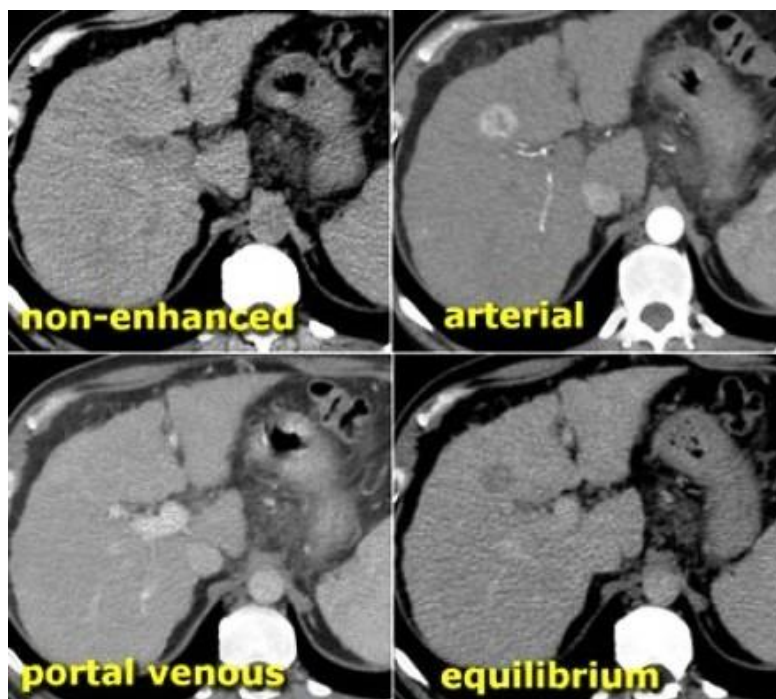


Figure 6. Triphasic CT scan showing enhancement of the lesion (HCC) in the arterial phase & wash-out of contrast in the venous portal phase.

Treatment

Treatment of 1^{ry} MTs of the liver is either curative or palliative:

Curative Treatment Options	Palliative Treatment Options
1. Hepatic resection: For patients with Child A where resection is done with adequate free margin & adequate residual liver volume. 2. Liver transplantation (OLTx, LDLT) for patients within Milan Criteria, regardless of the Child class. 3. Local ablation for solitary tumors, < 3 cm.	1. TACE (Trans-Arterial Chemo-Embolization) (the best palliative option) 2. Local radiotherapy 3. Targeted chemotherapy (Sorafenib) 4. Systemic chemotherapy 5. Ligation of hepatic artery 6. Percutaneous ethanol injection

Operable Cases

The ideal treatment for a tumor localized to one liver lobe is to resect that lobe. The usual operation for such situation is either right hemi-hepatectomy (resection of segments 5-8) or left hemi-hepatectomy (resection of segments 2, 3 and 4) (Figure 7).

The presence of cirrhosis in most cases poses some problems due to:

1. Bleeding tendency.
2. Poor function of the remaining 1/2 of liver
3. ↓ capacity of liver cell regeneration.

For these reasons, hepatic resection is reserved for patients with fair liver functions.

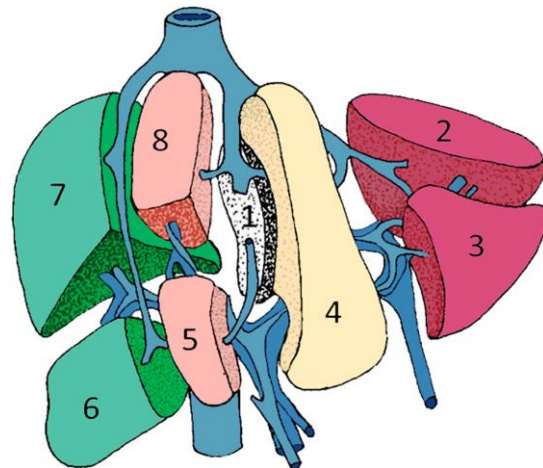


Figure 7. Segments of the liver. Hepatic resection is named according to the segments resected.

Inoperable Cases

The following palliative options are available

1. *Trans-Arterial Chemo-Embolisation (TACE)* combines the benefits of inducing tumor ischemia and those of selective chemotherapy. The chemotherapy agent is loaded on a particulate material like *gelfoam* and is injected by angiographic techniques into the HA. The artery is blocked → tumor ischemia and the drug is selectively delivered to its target.
2. *Systemic chemotherapy.*
3. *Intra-arterial selective chemotherapy* via a catheter inserted in the HA.
4. *Ligation of HA:* As the blood supply of the tumor is exclusively through the HA, its ligation produces some regression.
5. *Percutaneous 90% ethanol injection* (under US guidance) → tumor necrosis.

Management According to Staging

The most accepted staging of HCC that links clinical data and therapeutic options is: The **Barcelona Clinic Liver Cancer (BCLC)** (Figure 8)

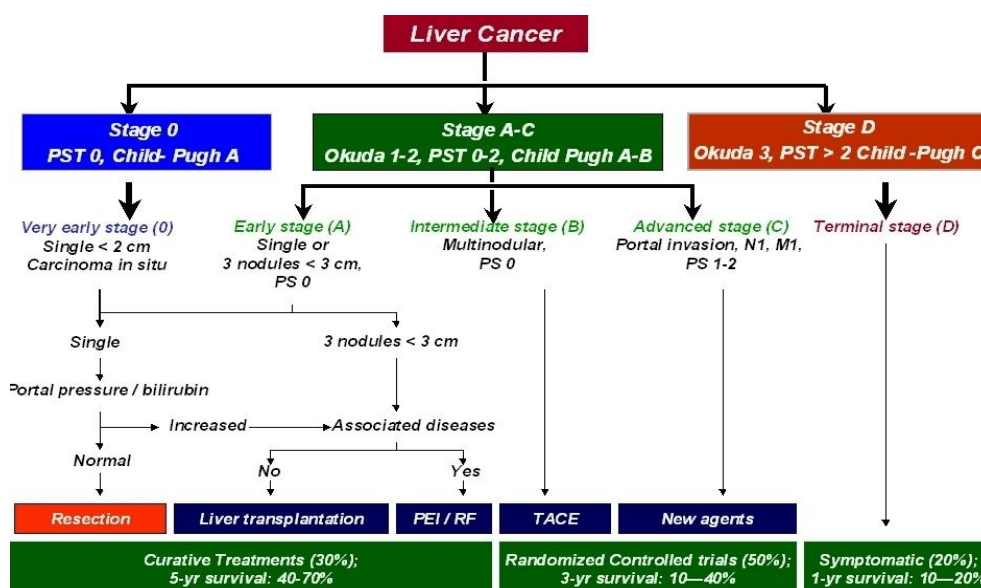


Figure 8: The Barcelona Clinic Liver Cancer (BCLC) staging of HCC

- **Stage 0:** Patients with very early HCC are optimal candidates for resection.
- **Stage A:** Patients with early HCC are candidates for radical therapies (resection and ablation, liver transplantation; or percutaneous treatments).
- **Stage B:** Patients with intermediate HCC may benefit from chemo-embolization.
- **Stage C:** Patients with advanced HCC may receive new agents in the setting of a RCT.
- **Stage D:** Patients with end-stage disease will receive symptomatic treatment.

B. LIVER METASTASES

Incidence

- Liver Metastases are 20 times more common than 1^{ry} malignancies.

Sources

Metastases Can Reach the Liver Through:

1. Portal circulation (commonest rout): From carcinoma of colon, rectum, pancreas or stomach.
2. Hepatic artery: Carcinomas of the lung, breast, kidney, uterus or ovaries are common examples.
3. Lymphatics.
4. Direct spread from tumors of the gall bladder, stomach or colon.

Metastatic Hepatic Focal Lesions

1. Colo-rectal hepatic metastases.
2. Neuro-endocrine hepatic metastases.
3. Non-colorectal, non-neuroendocrine hepatic metastases.

Pathology

- Metastases may occur in the same time of discovery of the 1^{ry} tumor (*synchronous*) or months or years after resection of the 1^{ry} (*metachronous*).
- Liver metastatic nodules are usually multiple, white, and umbilicated because of central necrosis (Figure 9).
- They are usually adenocarcinomas.
- Over 90% of patients have tumor deposits in other organs.



Figure 9. Liver secondaries (metastases)

Clinical Features

- In addition to the manifestations of the 1^{ry} MT, the patient suffers from weight loss, fatigue and anorexia, right upper abdominal pain, jaundice, ascites and/or palpable hepatomegaly.

Investigations

- Laboratory tests usually reveal anemia + ↑ serum alkaline phosphatase and bilirubin.
- Imaging by US, CT scan (Figure 10), or MRI shows the number and sites of metastases, which affects the line of treatment.

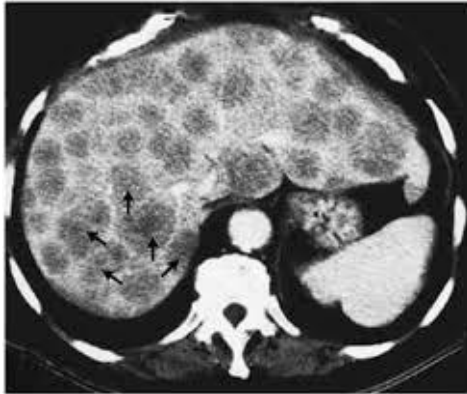


Figure 10. CT scan showing multiple metastases in the liver

Treatment

- The presence of liver metastases indicates an advanced inoperable tumor.
- The treatment in such cases, usually by chemotherapy, is considered palliative.
- A few selected cases of metastatic colorectal cancer, if localized to one segment, can be treated by liver resection aiming at cure.

Chemotherapy

1. *Systemic chemotherapy* is administered in the presence of other organ involvement.
2. *Selective intra-arterial chemotherapy* via a catheter surgically inserted in the HA, is indicated when the liver is the only organ with metastases. This technique reduces the systemic side effects of anti-cancer agents & allows delivery of higher doses to the liver.

Liver Resection

- The prerequisites include:
 1. A completely resectable colorectal cancer with no apparent residual tumor.
 2. No extra-hepatic metastases are found after a thorough search.
 3. Liver metastases that are resectable with a safety margin:
- A solitary metastatic nodule → *wedge resection*.
- Multiple metastases confined to one surgical lobe → right or left hemi-hepatectomy.
- Liver resection is done in the same session of the colorectal operation, or later if the liver deposits appear in the course of the patient's follow-up. Cure rate = 25%.

Prognosis

- In patients with isolated hepatic metastases, the extent of liver disease is the prime determinant of survival, and when left untreated, survival is measured in *months*.
- Surgery is safe and potentially curative in treatment of colo-rectal metastases to the liver.
- The current 5-year survival after a margin-negative hepatic resection is 40%, while the 10-year survival approaches 20% only.

Organ Transplantation and Liver Transplantation

TERMS AND TYPES OF TRANSPLANTATION

Organ transplantation

- It is a surgical procedure in which a failing organ in a recipient is replaced by a functioning one from a donor with matching tissue and blood compatibility.

Donor

- A donor is the healthy individual who accepts to donate his tissue to the diseased patient after receiving proper consent without any coercion.

Recipient

- A recipient is the patient who has a failing organ function that endangers his life and is willing to receive the new functioning organ from the donor.

Auto-transplantation

- It is the type of transplantation where tissues (auto-grafts) are transferred within the same person e.g. skin grafts and vein grafts.

Allo-transplantation

- It is the transplant of an organ or tissue (allograft) between two genetically non-identical members belonging to the same species. Most human tissue and organ transplants are allografts (from one human to another). Such genetic differences between the organ donor and the recipient makes it easy for the recipient's immune system to identify the transplanted organ as a foreign body and starts to act against it to attempt destroying it, which is known as transplant *rejection*.

Isograft

- It is the transplantation of organs or tissue between two genetically identical individuals (such as identical twins). Iso-grafts differ from allografts in that they do *not* trigger the recipient's immune response since it is transplanted from a genetically identical person.

Xenograft and Xeno-transplantation

- It is the transplantation of organs or tissue between members of two different species e.g. porcine heart valve transplant.

Orthotopic transplantation

- It entails placement of the graft in its normal anatomical site.

Heterotopic transplantation

- It entails placement of a graft in a site different from where it is normally located.

Split transplantation

- It is the type of transplantation in which an allograft procured from a deceased donor can be split to benefit 2 recipients, such as liver allograft, which can be split into right lobe and left lobe or right tri-segment lobe and left lateral lobe to fit into an adult recipient and a pediatric recipient, respectively.

Domino transplantation

- It is the type of transplantation where a healthy donor donates his organ to a diseased recipient after taking out his failing organ. This failing organ in return can be used in another recipient who is in a grave situation and who can benefit from such failing organ for some time until a healthy donor is available to offer him his organ. This type is suitable only in liver grafts with amyloidosis, or lungs with cystic fibrosis.

BASIC IMMUNOLOGY OF TRANSPLANTATION

- Donor allografts trigger the recipient's immune response owing to antigenic differences between them, which promote antibody formation and consequently lead to graft rejection. Accordingly, prior to transplantation, histocompatibility matching between donor antigens and recipient antigens is required to minimize rejection.
- The most 2 important antigen groups that should be tested for compatibility between donor and recipient are:
 1. ABO blood group antigens
 2. Human leukocyte antigens (HLA)
- Major organs liable for transplantation:
 1. Heart
 2. Lung
 3. Liver
 4. Kidney
 5. Pancreas
 6. Small bowel
- Tissues liable for transplantation:
 1. Skin
 2. Cornea
 3. Pancreatic islets
 4. Bone marrow
 5. Tendon
 6. Cornea
 7. Heart valve
 8. Bone
 9. Veins

Types of Rejection

Hyper-acute Rejection

- This type of rejection occurs *immediately* within few hours following transplantation.
- It usually occurs due to the presence of *preformed antibodies* in recipient's blood against HLA antigens expressed in the donor's allograft.
- It also occurs with organ transplantation from an ABO-incompatible donor.

Acute Rejection

- It usually occurs during the first few days up to 6 months after transplantation.
- It is usually *cell-mediated* immune reaction (by T-cell lymphocytes), but may also involve humoral immune reaction (antibody-mediated), or both.

Chronic Rejection

- It usually occurs more than 6 months following transplantation.
- It may be immunological or non-immunological.
- It is the most common cause of allograft failure.

ORGAN PROCUREMENT

Organ procurement can be performed from:

1. *Living donors* who can donate organs as one kidney, right or left lobe of the liver, or a lobe of the lung.
2. *Deceased donors* who can donate a whole organ and may be classified into:
 - a) Donation after donor brainstem death (DBD).
 - b) Donation after donor circulatory death (DCD).

INDICATIONS OF LIVER TRANSPLANTATION

Liver transplantation is indicated for patients suffering from irreversible decompensated end-stage liver disease (ESLD) due to any of the following factors:

Acute Liver Failure

1. Acute viral hepatitis (HAV, HBV, HEV).
2. Drug- or toxin-induced hepatotoxicity (e.g. Acetaminophen hepatic toxicity).
3. Autoimmune hepatitis.
4. Wilson's disease.

Chronic Liver Failure

1. Chronic viral hepatitis (HCV, HBV)
2. Alcoholic liver disease.
3. Non-alcoholic fatty liver disease and steato-hepatitis
4. Autoimmune hepatitis
5. Cryptogenic liver cirrhosis

Malignant Diseases of the Liver

1. Hepatocellular carcinoma (HCC), within *Milan criteria* (single lesion within 5 cm or up to 3 lesions, none of them exceeding 3 cm in diameter).
2. Carcinoid tumor
3. Cholangio-carcinoma
4. Hemangio-endothelioma

Metabolic Liver Diseases

1. Wilson's disease
2. Hereditary hemochromatosis
3. Alpha-1 antitrypsin deficiency
4. Glycogen storage disease

Vascular Diseases

1. Budd-Chiari syndrome
2. Veno-occlusive diseases of the liver

Cholestatic Liver Diseases

1. Primary biliary cholangitis (PBC)
2. Primary sclerosing cholangitis (PSC)
3. Biliary atresia
4. Alagille syndrome (a genetic disorder that affects primarily the liver and heart)

Others

1. Adult polycystic liver disease
2. Liver trauma
3. Caroli disease
4. Amyloidosis

LIVING DONORS

Investigations for Living Donor Assessment

Laboratory Tests

1. Complete blood count (CBC)
2. Liver function tests (LFTs)
 - Total and direct bilirubin
 - Serum albumin and prothrombin
 - Liver enzymes: AST, ALT
 - Cholestatic enzymes: ALP, GGT
3. Coagulation profile: INR, PT, PTT
4. Renal function tests: BUN, serum creatinine
5. Hepatitis profile for HBV, HCV, HAV, HEV, HDV
6. Chronic infections as CMV, EBV
7. Urinalysis for drug abuse

Cardiac Assessment

1. Electrocardiography (ECG)
2. Echo-cardiography
3. Cardiac catheterization.

Imaging Studies

1. Chest X-ray
2. Abdomino-pelvic ultrasound (US)
3. Duplex US of the portal vein and hepatic artery
4. Triphasic abdominal CT with volumetry and/or MRI
5. CT angiography

Ideal Liver Transplant Living Donor

Criteria of an ideal liver transplant living donor:

1. Age: 8-45 years
2. BMI < 30
3. No previous upper abdominal surgery
4. No history of any chronic diseases as hypertension or diabetes mellitus.
5. No evidence of drug abuse or smoking
6. No evidence of HCV, HBV, or bilharziasis
7. No coercion to donation
8. Related to the recipient
9. Absence of any financial benefit

CONTRAINDICATIONS OF LIVER TRANSPLANTATION

Absolute Contraindications

1. Severe cardiopulmonary disease.
2. Extra-hepatic malignancy (oncological criteria for cure not met).
3. Active alcohol/substance abuse.
4. Acute alcoholic hepatitis.
5. Active infection/uncontrolled sepsis.
6. Lack of psychosocial support or inability to comply with medical treatment.
7. Brain death.

Relative Contraindications

1. Advanced age.
2. Acquired immune deficiency syndrome (AIDS).
3. Cholangio-carcinoma.
4. HCC exceeding the transplantable criteria.
5. Diffuse portal vein thrombosis.

IMMUNO-SUPPRESSION AFTER LIVER TRANSPLANTATION

Steroid-based triple therapy is followed in the form of:

1. Corticosteroids.
2. Anti-proliferative drugs such as mycophenolate.
3. Calcineurin inhibitors such as cyclosporine A or tacrolimus.

COMPLICATIONS OF LIVER TRANSPLANTATION

Complications of liver transplantation include the following:

1. *Biliary complications*: stricture – leak - stone formation
2. *Vascular complications*: hepatic artery thrombosis or stenosis - Portal vein thrombosis - Hepatic vein or inferior vena cava (IVC) occlusion or thrombosis.
3. *Parenchymal complications*: Recurrent disease (as HCV, HCC) - acute or chronic liver allograft rejection
4. *Opportunistic infections*
5. *Immunosuppressive drugs-induced complications*

Abdominal Trauma

TOPOGRAPHIC ANATOMY

Definitions (Figure 1)

- *Thoraco-abdominal area*: Transverse nipple line to costal margin
- *Anterior abdomen*: Costal margin to groin crease to anterior axillary lines bilaterally
- *Flank area*: Anterior axillary line to posterior axillary line, costal margin to iliac crests
- *Back*: Medial to posterior axillary lines, tip of scapula to iliac crests
- *Torso*: All the above



Figure 1.
Topographic anatomy of the abdomen

The Abdomen is More Than Just the Abdomen

- Abdominal trauma may involve the following (Figure 2):
 1. *Intra-peritoneal cavity*
 2. *Retro-peritoneal cavity and pelvis*
 3. *Thoraco-abdominal injuries*
 4. *Cardiac box injuries* (heart and great vessels)
 5. *Diaphragm and bladder* (innocent by-standers)

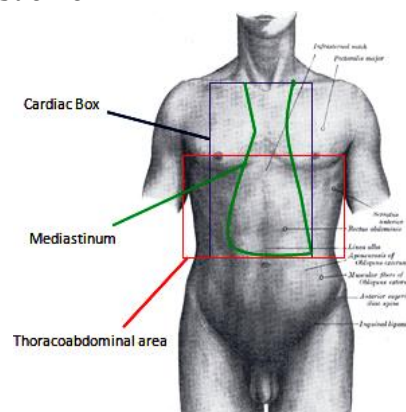


Figure 2. Parts of the abdomen that may be involved in case of abdominal trauma

CLASSIFICATION

According to Etiology

1. *Penetrating Abdominal Trauma (PAT)*
 - Stab wounds (75%)
 - Gunshot wounds (high-velocity, low-velocity) (25%)
2. *Blunt Abdominal Trauma (BAT)*
 - High-energy transfer (Motor Vehicle Accident – MVA) (75%)
 - Low-energy transfer (fall, fight) (25%)
3. *Mixed Pattern: Explosions*

According to Pathogenesis or Mechanism of Injury (MOI)

1. *Blunt Abdominal Trauma (BAT)*

- Examples: Road traffic accident (RTA), run-over, falls, blows, etc.
- Multiple organ damage is quite often.
- Most commonly injured organs: Small intestine > colon > liver
- Pressure effect → burst injury,
- Crushing effect → crush injury
- Acceleration and deceleration effect → shear injury.

2. *Penetrating Abdominal Trauma (PAT)*

Stab Wounds

- Examples: Knives, ice picks, pens, coat hangers, broken bottles, etc.
- Multiple organ injuries are less often than with BAT.
- Most commonly injured organs: Liver, small bowel & spleen.
- Direct effect.

Gunshot Wounds

- Examples: Bullets and pellets
- Often multiple organ injuries, bowel perforations
- Most commonly injured organs: Small bowel, colon & liver
- Direct effect
- Kinetic energy (*cavitation*).

According to Grading

AAST Concept (American Association for the Surgery of Trauma)

- Grading of abdominal injuries is done according to the AAST concept according to certain *CT-based criteria*, mainly relying on the following:
 1. Presence of hematoma
 2. Presence of laceration
 3. Presence of vascular injury
 4. Effect on the collecting system.
- Accordingly, solid abdominal organs (spleen, liver, kidneys and pancreas) can be classified (graded) as follows:

Grading of the Splenic Injury (Figure 3)

Grade	Description	
Grade I	Hematoma:	Subcapsular, non-expanding hematoma (< 10% surface area)
	Laceration:	Capsular tear, non-bleeding, parenchyma involved <1cm deep
Grade II	Hematoma:	Subcapsular, nonexpanding hematoma (10-50% surface area); nonexpanding, intra-parenchymal hematoma (<2cm in diameter)
	Laceration	Bleeding capsular tear or parenchymal laceration 1-3 cm deep without trabecular vessel involvement
Grade III	Hematoma	Subcapsular hematoma (> 50% surface area or expanding); ruptured subcapsular hematoma with active bleeding; or intra-parenchymal hematoma (> 2 cm in diameter or expanding)
	Laceration	Parenchymal laceration > 3 cm deep or involving trabecular vessels
Grade IV	Hematoma	Ruptured intra-parenchymal hematoma with active bleeding
	Laceration	Involving segmental or hilar vessels producing major (> 25% splenic volume) Devascularization
Grade V	Laceration	Completely shattered or avulsed spleen
	Vascular	Hilar vascular injury that devascularizes the entire spleen

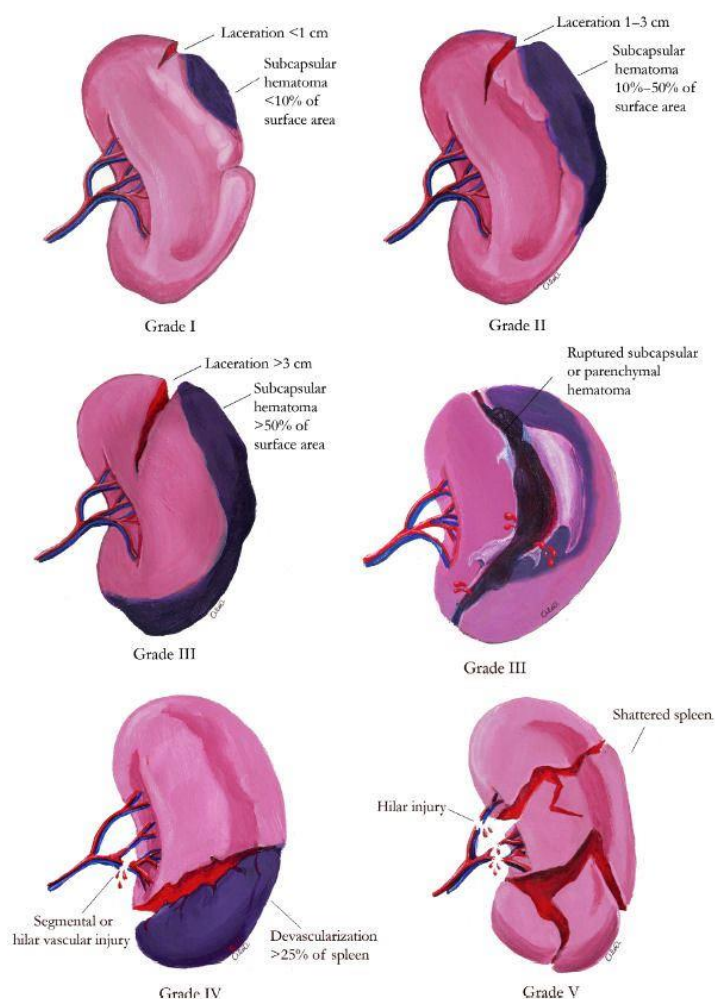


Figure 3. Grading of splenic injury

Easy way to remember

Grade 1 is < 1 cm

Grade 2 is about 2 cm

Grade 3 is > 3 cm

Grade 4 is > 10 cm

Grade 5 corresponds to total devascularization or maceration

Grading of the Liver Injury (Figure 4)

Hepatic CT Injury Grading Scale	
Grade I	Laceration(s) < 1 cm deep Subcapsular hematoma < 1cm diameter
Grade II	Laceration(s) 1-3 cm deep Subcapsular or central hematoma 1-3cm diam
Grade III	Laceration(s) 3-10 cm deep Subcapsular or central hematoma 3-10 cm diam
Grade IV	Laceration(s) > 10 cm deep Subcapsular or central hematoma > 10cm diam Lobar maceration or devascularization
Grade V	Bilobar tissue maceration or devascularization

Easy way to remember

Grade 1 is < 1 cm

Grade 2 is about 2 cm

Grade 3 is > 3 cm

Grade 4 is > 10 cm, or unilobar maceration

Grade 5 is bilobar maceration, venous injury

Grade 6 is avulsion

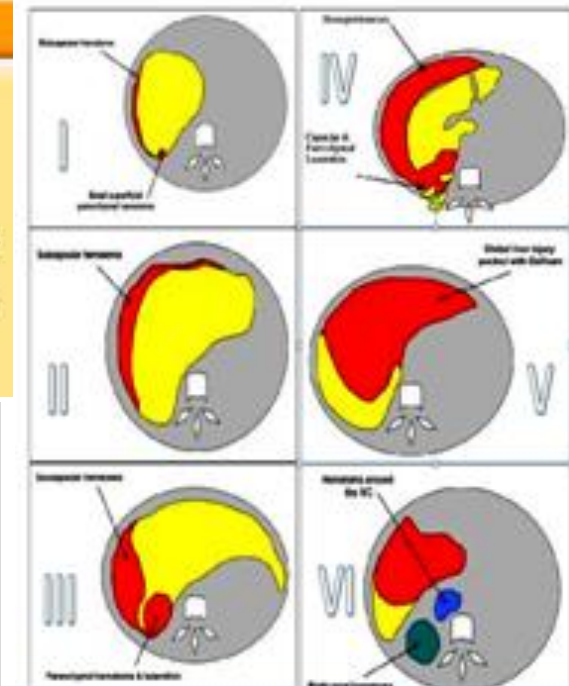


Figure 4. Grading of hepatic injury

Grading of the Renal Injury (Figure 5)

Grade	Extent of Renal Injury
1	<i>Contusion:</i> Microscopic or gross hematuria, no injury with any imaging <i>Hematoma:</i> Subcapsular with no parenchymal laceration
2	Non-expanding peri-renal hematoma or cortical laceration < 1cm deep with no urinary extravasation.
3	Parenchymal laceration extending > 1cm into the cortex with no urinary extravasation.
4	Parenchymal laceration extending through the cortico-medullary junction & into the collecting system.
5	Multiple major lacerations resulting in shattered kidney or avulsion of the renal hilum that devascularizes the kidney.

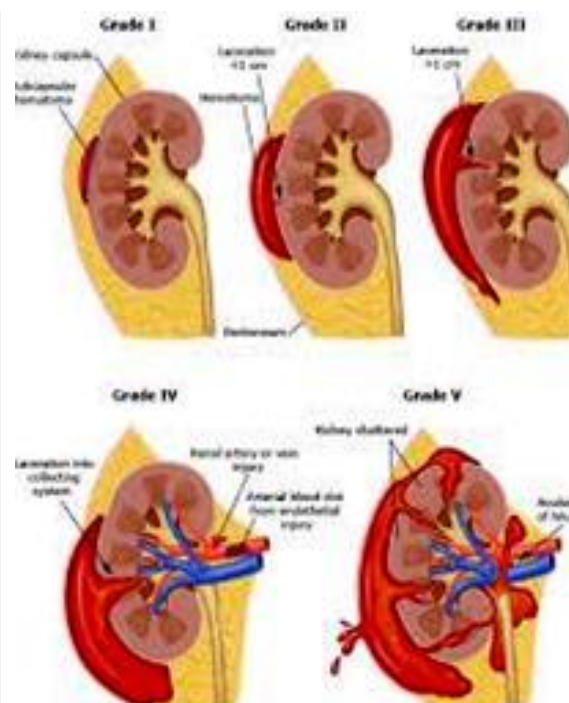


Figure 5. Grading of renal injury

Easy way to remember

- **Grade 1** is contusion or subcapsular hematoma only
- **Grade 2** is laceration < 1 cm without injury to collecting system
- **Grade 3** is laceration > 1 cm without injury to collecting system
- **Grade 4** is injury to collecting system or large laceration
- **Grade 5** is shattered or devascularized kidney

Grading of the Pancreatic Injury (Figure 6)

	Grade	Injury description
I	Haematoma	Minor contusion without ductal injury
	Laceration	Superficial laceration without ductal injury
II	Haematoma	Major contusion without ductal injury or tissue loss
	Laceration	Major laceration without ductal injury or tissue loss
III	Laceration	Distal transection or pancreatic parenchymal injury with ductal injury
IV	Laceration	Proximal transection or pancreatic parenchymal injury involving the ampulla
V	Laceration	Massive disruption of the pancreatic head

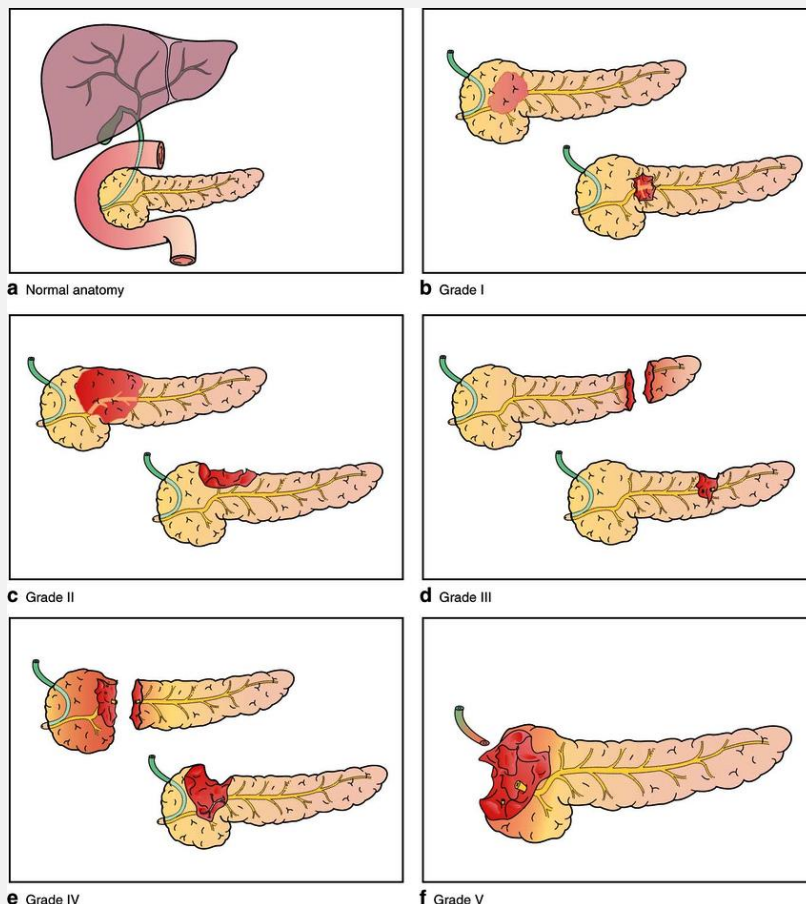


Figure 6.
Grading
of
pancreatic
injury

Physical Examination

Signs of Intra-Peritoneal Injury

- Hypovolemia
- Peritonism.
- Distention (pneumo-peritoneum, gastric dilation, or ileus)
- SC emphysema
- Entrance & exit wounds to determine path of injury.
- Gastrointestinal hemorrhage: Mouth or rectal
- Ecchymosis:
 1. Umbilicus (**Cullen's sign**) (Figure 7)
 2. Flanks (**Gray Turner's sign**) (Figure 8): Retro-peritoneal hemorrhage
 3. Back: Retroperitoneal hemorrhage
 4. Abdominal contusions – **Seat belt sign** (Figure 9),



Figure 7. Cullen's sign: ecchymosis around the umbilicus



Figure 8. Grey-Turner's sign: ecchymosis around the umbilicus

Be cautious !

Clinical assessment is mandatory but is not enough because of.

- *Silent presentation*
Blunt trauma can be evolving.
- Limitations because of the degree of level of consciousness, distracting injury and spinal cord injury.



SEAT-BELT SIGN



Figure 9. Abdominal contusion: Characteristic Seat-belt sign

INVESTIGATIONS

Laboratory Investigations

- CBC (complete blood count): ↑ WBC (leucocytosis) - ↓ HCV (hematocrit value)
- ↑ CRP (C-reactive protein).
- LFTs (liver function tests) - ↑ lipase (pancreatic enzyme).

Imaging Studies

Plain X-Ray

A plain radiogram may show the following:

- *Fractures*: nearby visceral damage
- *Air*: Subcutaneous - Free intra-peritoneal air (Figure 10) - Retro-peritoneal air
- *Foreign bodies & missiles* (Figure 11)

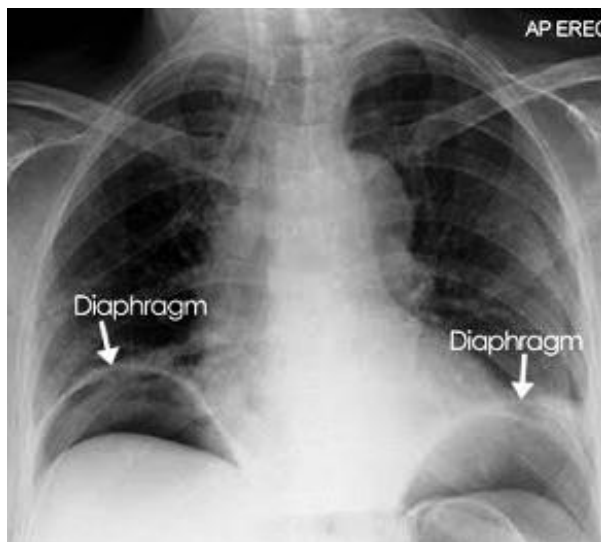


Figure 10. Plain X-ray abdomen showing air under diaphragm (free intra-peritoneal)



Figure 11. Plain X-ray abdomen showing foreign body (missile)

FAST (Focused Assessment with Sonography for Trauma)

- To diagnose free intra-peritoneal blood after blunt trauma, 4 areas should be examined:
 1. Peri-hepatic and hepato-renal space (**Morrison's pouch**) (Figure 12)
 2. Peri-splenic (Figure 13)
 3. Pelvis (**Douglas pouch**/recto-vesical pouch) (Figure 14)
 4. Pericardium (sub-xiphoid)
- Sensitivity 60-95% for detecting 100 mL - 500 mL of fluid
- **Extended FAST (E-FAST)**: Add *thoracic windows* to look for pneumothorax (Figure 15)
- **Advantages of FAST**
 1. Portable
 2. Fast (<5 minutes),
 3. No radiation or contrast material injected
 4. Less expensive when compared to CT scan

- *Disadvantages of FAST*

1. Operator-dependent
2. Limited by obesity, substantial bowel gas and subcutaneous (SC) air.
3. Not good for assessment of the retro-peritoneum, or detection of diaphragmatic defects.
4. Not suitable for solid parenchymal damage
5. Cannot distinguish blood from other fluid collections.

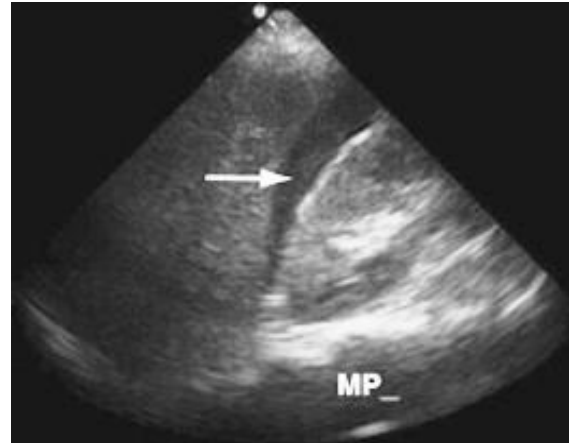


Figure 12. FAST of the hepato-renal space (Morrison pouch) for injuries of liver and kidney

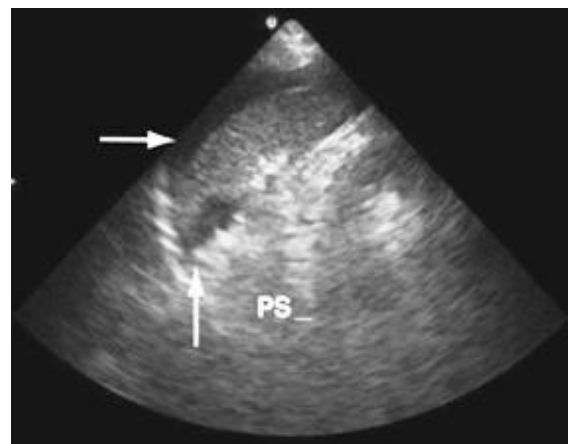


Figure 13. FAST of the peri-splenic region for assessment of injuries of the spleen



Figure 14. FAST of the recto-vesical pouch (Douglas pouch)

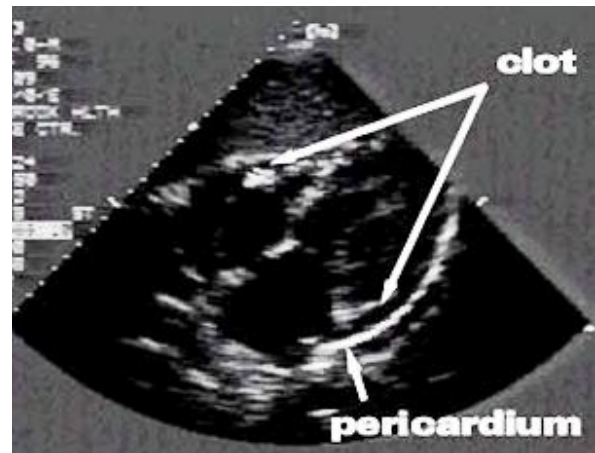


Figure 15. FAST of the pericardium (sub-xiphoid area)

CT scan

- *Advantages*
 1. Accurate for solid organ lesions and intra-peritoneal hemorrhage (Figure 16 A-D)
 2. Can guide non-operative management of solid organ damage
- *Disadvantages*
 1. Expensive.
 2. Radiation & contrast exposure.
 3. Less sensitive for injury of the pancreas, diaphragm & mesentery.

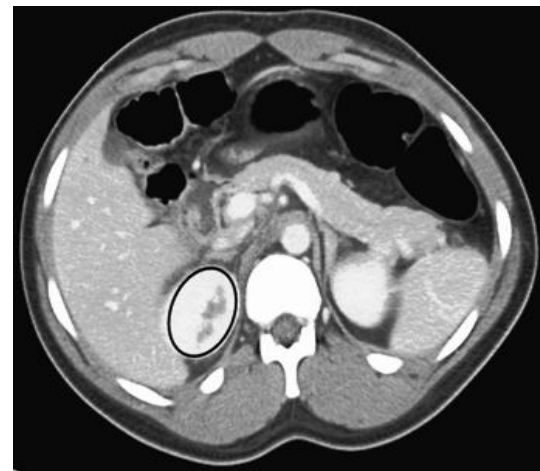
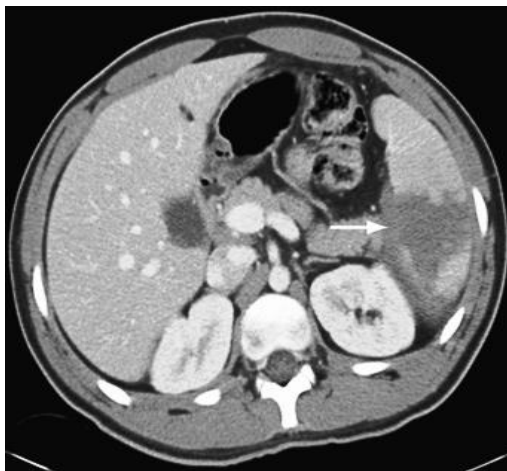


Figure 16. CT scan of the abdomen showing solid organ & intra-peritoneal lesions (injuries)

Angiography

- To embolize bleeding vessels or solid visceral hemorrhage from blunt abdominal trauma (BAT) in an *unstable* patient
- Rarely, for diagnosing intra-peritoneal & retro-peritoneal hemorrhage after penetrating abdominal trauma (PAT)

Diagnostic Peritoneal Lavage (DPL)

- *Procedure (under sterile conditions)*
 1. Attempt to aspirate free peritoneal blood: >10 mL is +ve for intra-peritoneal injury
 2. Insert lavage catheter.
 3. Lavage peritoneal cavity with saline
- *Positive test*
 1. In blunt trauma, or stab wound to anterior, flank, or back: RBC count > 100,000/mm³
 2. In lower chest stab wounds or GSW: RBC count > 5,000-10,000/mm³
- *DPL has been largely replaced by FAST and CT scan*
- *Limited indications*
 1. In blunt trauma, used to triage the patient who is hemodynamically unstable and has multiple injuries with an equivocal FAST examination
 2. In stab wounds, for immediate diagnosis of hemo-peritoneum, determination of intra-peritoneal organ injury & detection of isolated diaphragm injury

Summary of Assessment According to Region

Abdomen	
Intra-peritoneal cavity	Retro-peritoneal cavity & pelvis
- Imaging: FAST - CT scan - DPL - Exploratory laparoscopy - Exploratory laparotomy	- Pelvic X-ray - CT scan - Exploratory laparotomy
Thorax (Thoraco-Abdominal Injuries)	Heart & Great Vessels (Cardiac Box Injuries)
- Chest X-Ray	- Cardiac FAST - Chest X-Ray
Diaphragm and Bladder (innocent bystanders)	
- Diagnostic laparoscopy - CT cystogram	

TREATMENT

Management of BAT (Algorithm) (Figure 17)

Non-operative Management - Indication Criteria

- Reliable FAST - Good quality CT scans - Experienced radiologist
- Intensive care (ICU) setting available
- Hemodynamically stable victim
- Minor-Grade Injury
- Absence of active hemorrhage
- Absence of peritoneal signs

Operative Management - Indication Criteria

- Lack of reliable FAST or good quality CT scans, or experienced radiologist
- Lack of Intensive care (ICU) setting
- Hemodynamically unstable victim
- High or increasing packed RBCs transfusion requirement.
- High-Grade Injury
- Active hemorrhage
- Peritonism

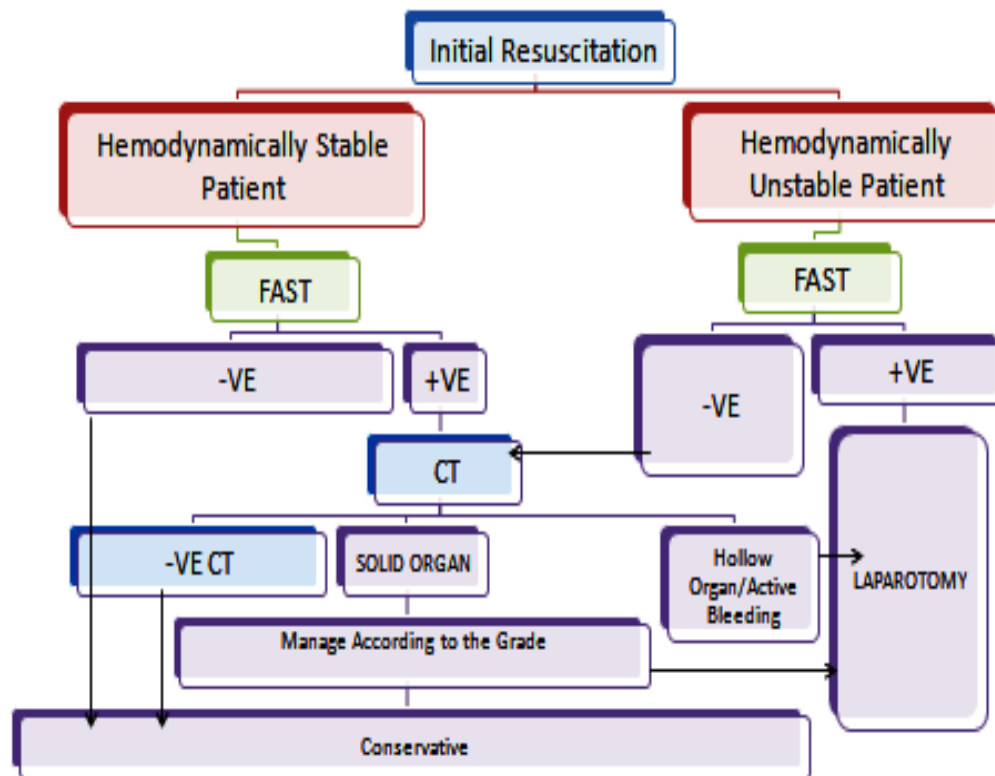


Figure 17. Algorithm for Management of BAT

Management of PAT (Algorithm) (Figure 18)

Local Wound Exploration

Aims

- To determine the depth of penetration in case of stab wounds.
- If peritoneum is violated, more diagnostic investigations should be ordered

Procedure

- Prepare the wound site
- Extend wound the wound under local anesthesia
- Carefully examine the wound – Avoid blind probing (to avoid injury of organs)

Indication

- Anterior abdominal stab wounds, less clear for other areas

Laparoscopy

Indication

- Most useful to evaluate penetrating wounds to thoraco-abdominal region in stable patient especially for injury of the diaphragm

Advantages

- Accurate: Sensitivity 87.5%, specificity 100%
- Organs can be repaired via the laparoscope e.g. diaphragm, solid viscera, stomach, small bowel.

Disadvantages

- Poor sensitivity for hollow visceral injury & the retro-peritoneum
- Complications may occur from trocar misplacement.
- In case of diaphragmatic injury, pneumothorax may occur during insufflation

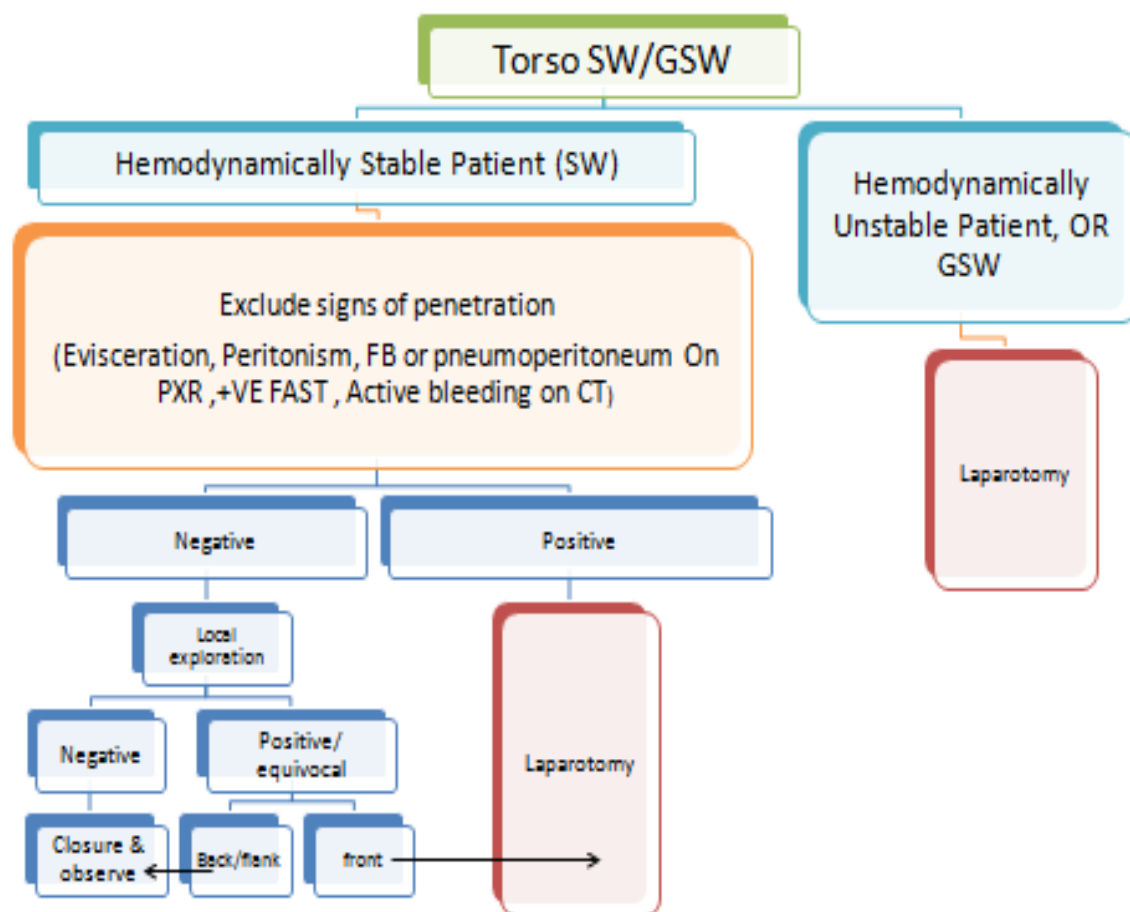


Figure 18: Algorithm for Management of PAT